



RFL-AMS-B

ANTENNA MEASUREMENT SYSTEM

EXPERIMENTAL MANUAL

....A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. ADVANCED DIGITAL COMMUNICATION is one of the important subject need to teach while learning electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

Company's Motto

- Light years ahead – refers to leadership.

We strive to be the best in what we do and maintain high standards in the areas of

• Design • Quality • Value • Delivery • Support

We are indeed light years ahead of the competition in this field. We guarantee our valued customers great satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete and comprehensive warranty.

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SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the power supply to the kit, be sure that the jumpers and the connecting chords are connected correctly, as per the experiment.
- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.
- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions' sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions' judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

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INTRODUCTION

Antennas are a fundamental component of modern communications systems. By definition, an antenna acts as a transducer between a guided wave in a transmission line and an electromagnetic wave in free space.

Why Antenna Measurements?

A graphical representation of the relative distribution of the radiated power in space is called a radiation pattern. The radiation pattern of the antenna is of principle concern when engineering a communications system. Let's assume that a signal needs to be sent from an antenna on the ground to a satellite in orbit. This would require a radiation pattern with the majority of its radiated power focused into orbit. If the antenna is not engineered to do so, contact cannot be established between the signal source and its target. There are many different ways to manipulate a radiation pattern to meet the demands of a specific task. These concepts are the principle focus of this lab assignment. Implementing this lab assignment, students will examine the radiation patterns of several antennas by hands on field testing.

Antennas demonstrate a property known as reciprocity that is an antenna will maintain the same characteristics regardless if it is transmitting or receiving. When a signal is fed into an antenna, the antenna will emit radiation distributed in space a certain way.

Motivation

Only the most fundamental antennas were chosen for this lab assignment. This allows us to see visually how the most common types of real-world antenna designs function.

Many antenna systems courses introduce theory and systems characteristics without a laboratory component. Since the development of antenna systems benefits from a physical understanding of wave propagation and competent measurement techniques, laboratory experiments should be an integral part of the lecture course.

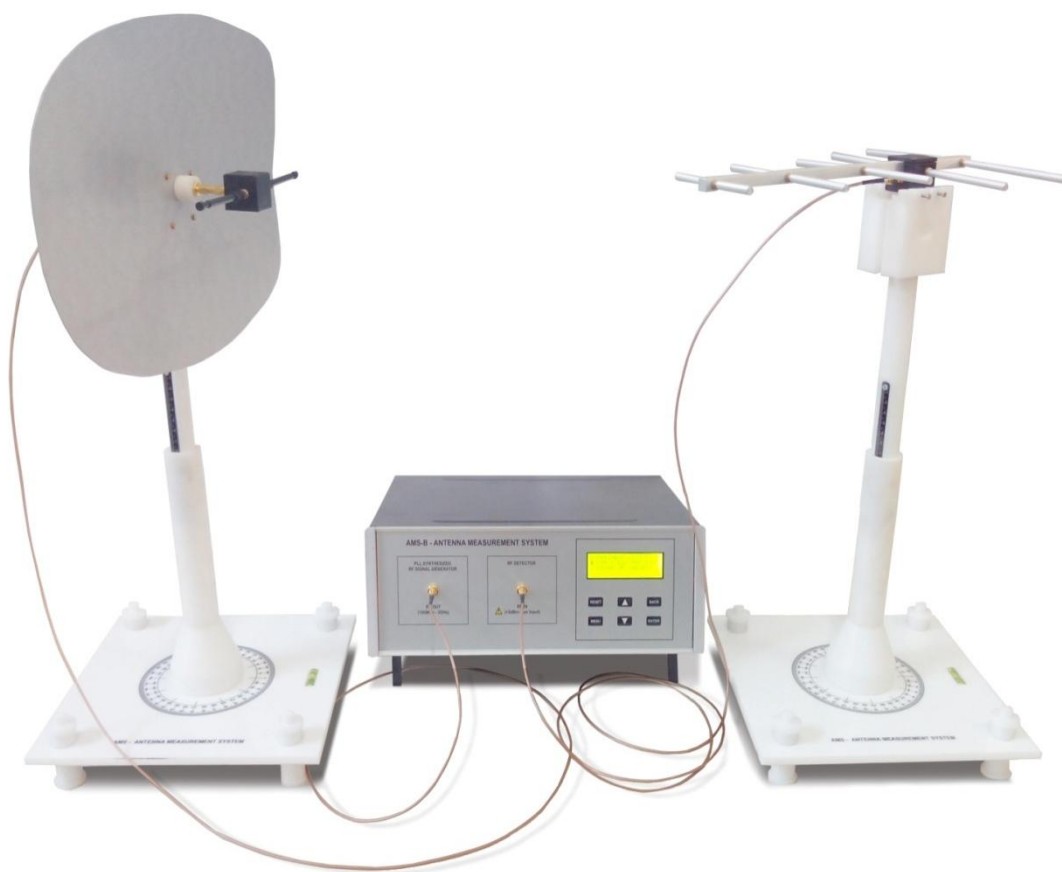
To address this issue, a complete laboratory course involving antenna measurements using Spectrum analyzer antenna measurement system has been developed



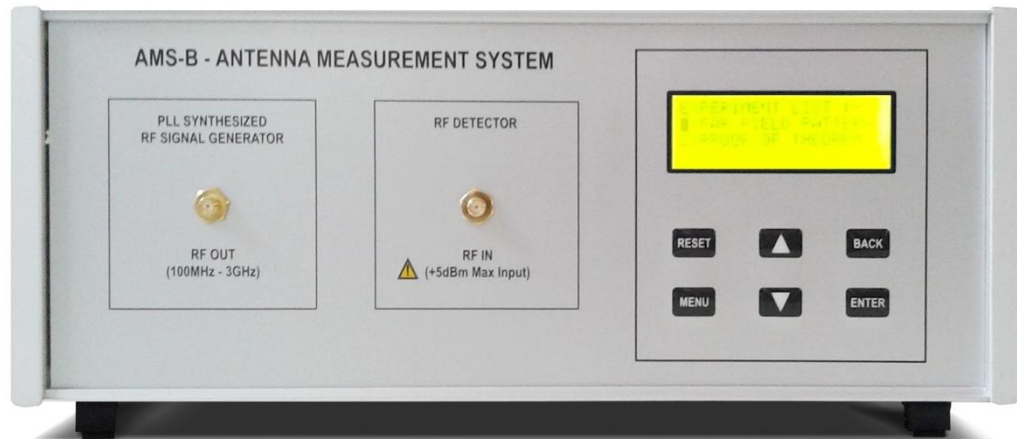
Features of Antenna Trainer System

- Covers UHF, L, S Band Wide frequency coverage up to 3 GHz
- PLL Synthesized Source and Detector working up to 3GHz with high dynamic range of power Transmission
- Customized selection of antenna from the list as per syllabus requirement
- Practical approach for Microstrip Antenna design covering concepts of size reduction, Bandwidth enhancement, stacked Multilayer configuration, impedance matching
- Non conductive and non magnetic Transmitter and Receiver pattern plotting software
- Single training system to teach all types of antenna measurement

FUNCTIONAL BLOCK



AMS-B FRONT PANEL DESCRIPTION



1. RF OUT

This is a PLL synthesized RF signal generator, the signal which is going to transmit will be available from this generator. The output frequency is from 100 MHz to 3GHz. Its output impedance is 50 ohms. The transmitting antenna can be connected to it using the SMA-SMA MALE cable provided. The power can be varying between -4dBm to +5dBm. An external attenuator is not needed.

2. RF IN

This is RF signal DETECTOR, where the RECEIVED signal from antenna should be connected. The RF IN terminal should connect to receiver antenna through cable. Its impedance is 50 ohms. The receiving antenna can be connected to it using the SMA-SMA MALE cable provided. The power can be vary between -4dBm to +5dBm. An external attenuator is not needed. Inbuilt software controlled attenuator is present.

3. LCD

This is a TFT LCD Display, used to display the MENU options as well as most of the selections done by the user i.e. frequency, rotation angle, received power in dBm, instructions and message is displayed.

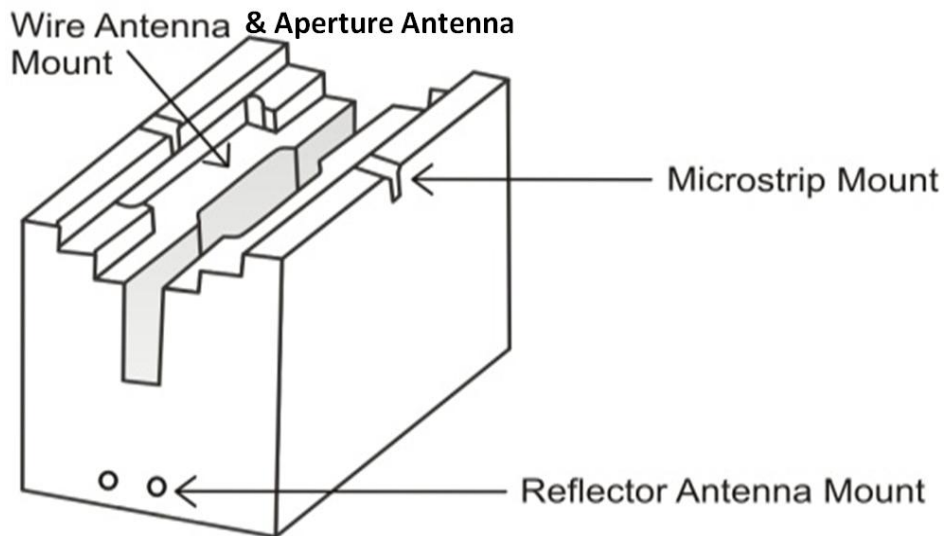
4. KEYBOARD

The keyboard is provided to give input to trainer.

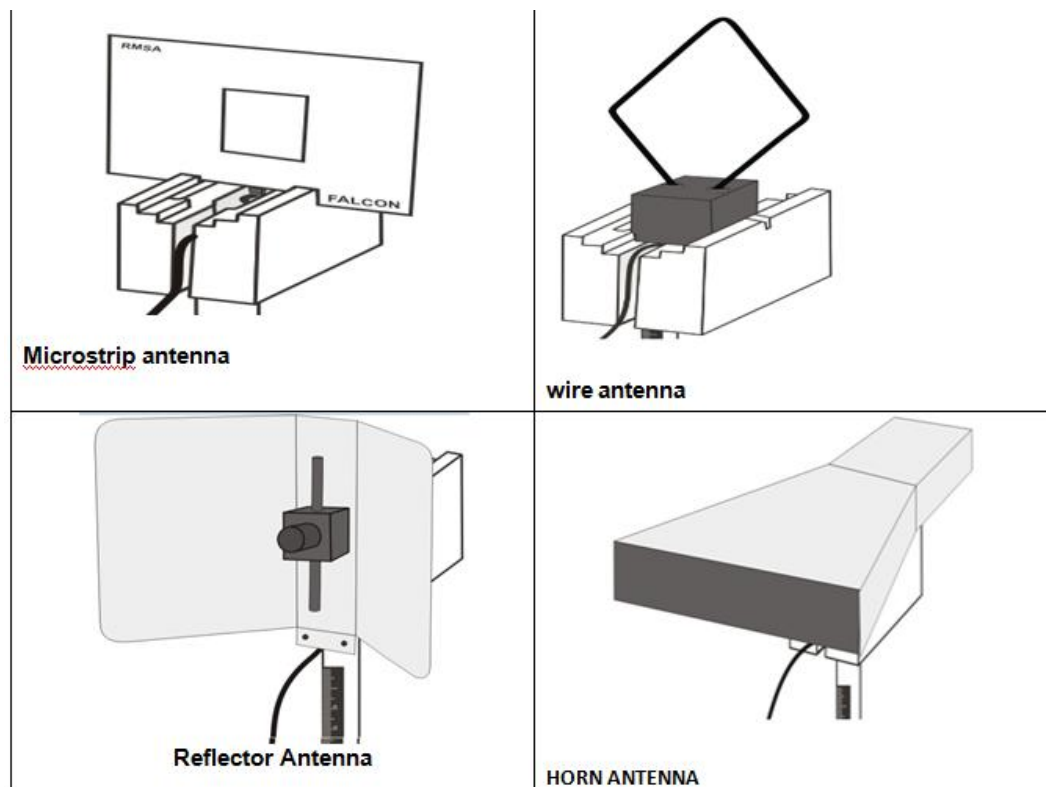
- MENU: by pressing this key the menu options are available.
- BACK: by pressing this key a user can move one step backward.
- ENTER: when a user wants to confirm its selection, ENTER key should be pressed.
- RESET: by pressing this key the trainer gets reset.

HORIZONTAL ASSEMBLY DESCRIPTION

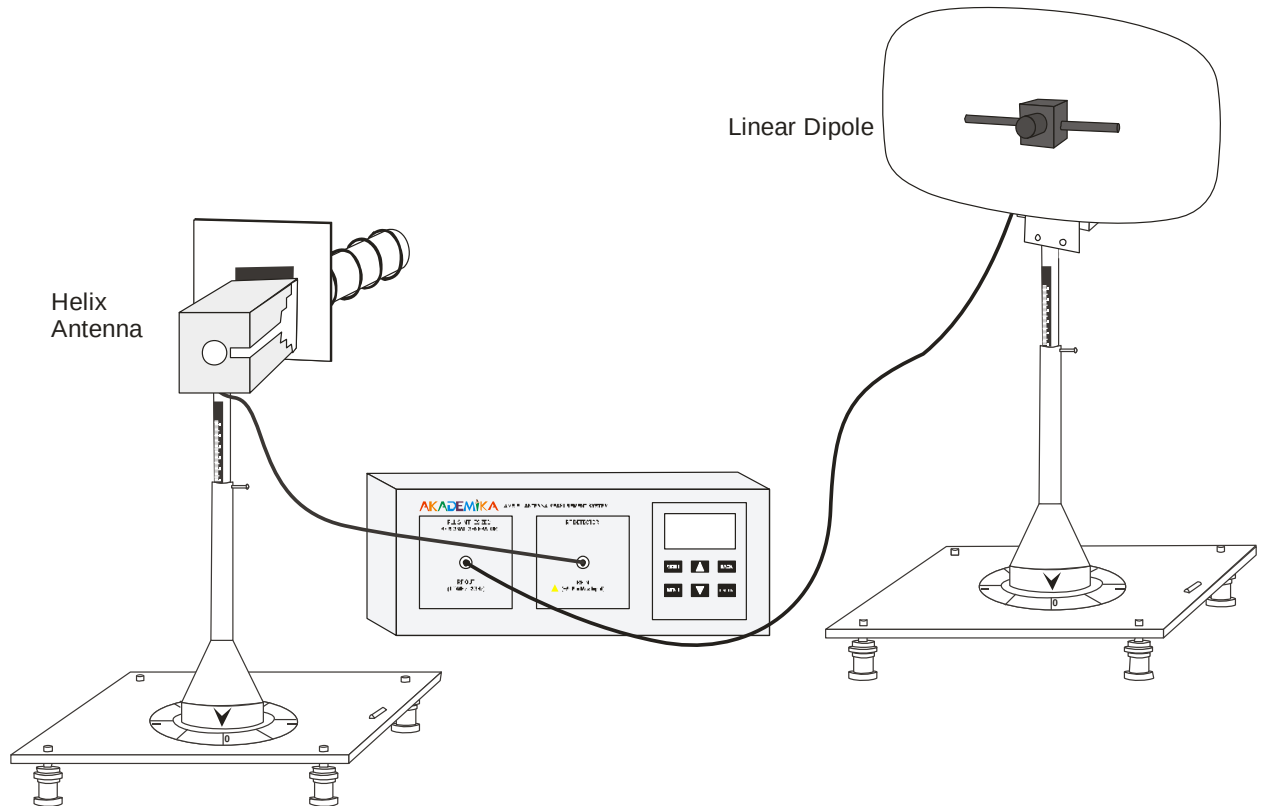
The below figure shows horizontal mount which is made up of polyxeamethalene polymer. This is universal mount to support transmitting and receiving antenna. Appropriate slots are provided to fit the antenna assembly. Press fit facility is available for Wire, aperture and planner antenna. The reflector antenna is required screw fitting.



The below examples are showing arrangement of antenna assembly with its appropriate fittings.

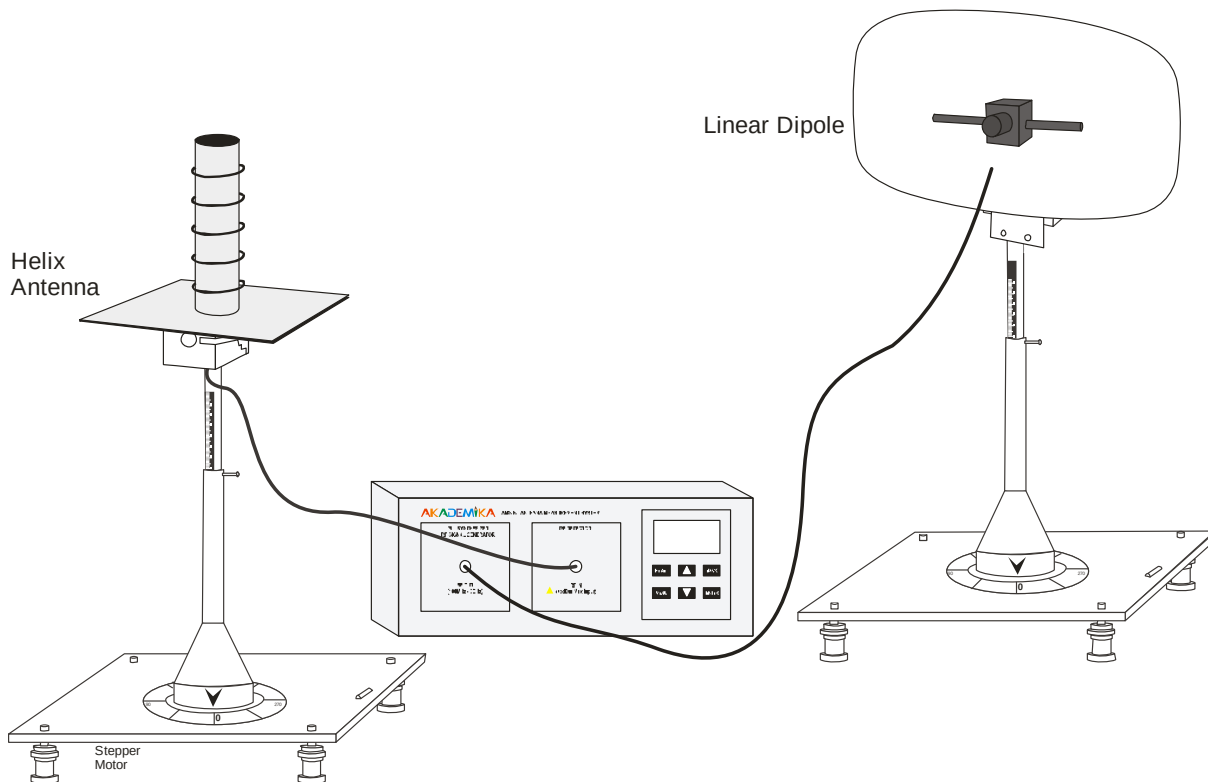


1. Attach four height adjuster to base plate
2. Check the spirit level; adjust the height of the supporter until the spirit bubbles come up in the center.
3. Press fit the conical mount to base plate. To reduce the friction the bearing is present.
4. Press fit Hollow rod to conical mount
5. Fit the horizontal mount to solid rod; rotate the mount clockwise so threads are fit with solid rod.
6. Insert the above assembly in hollow pipe

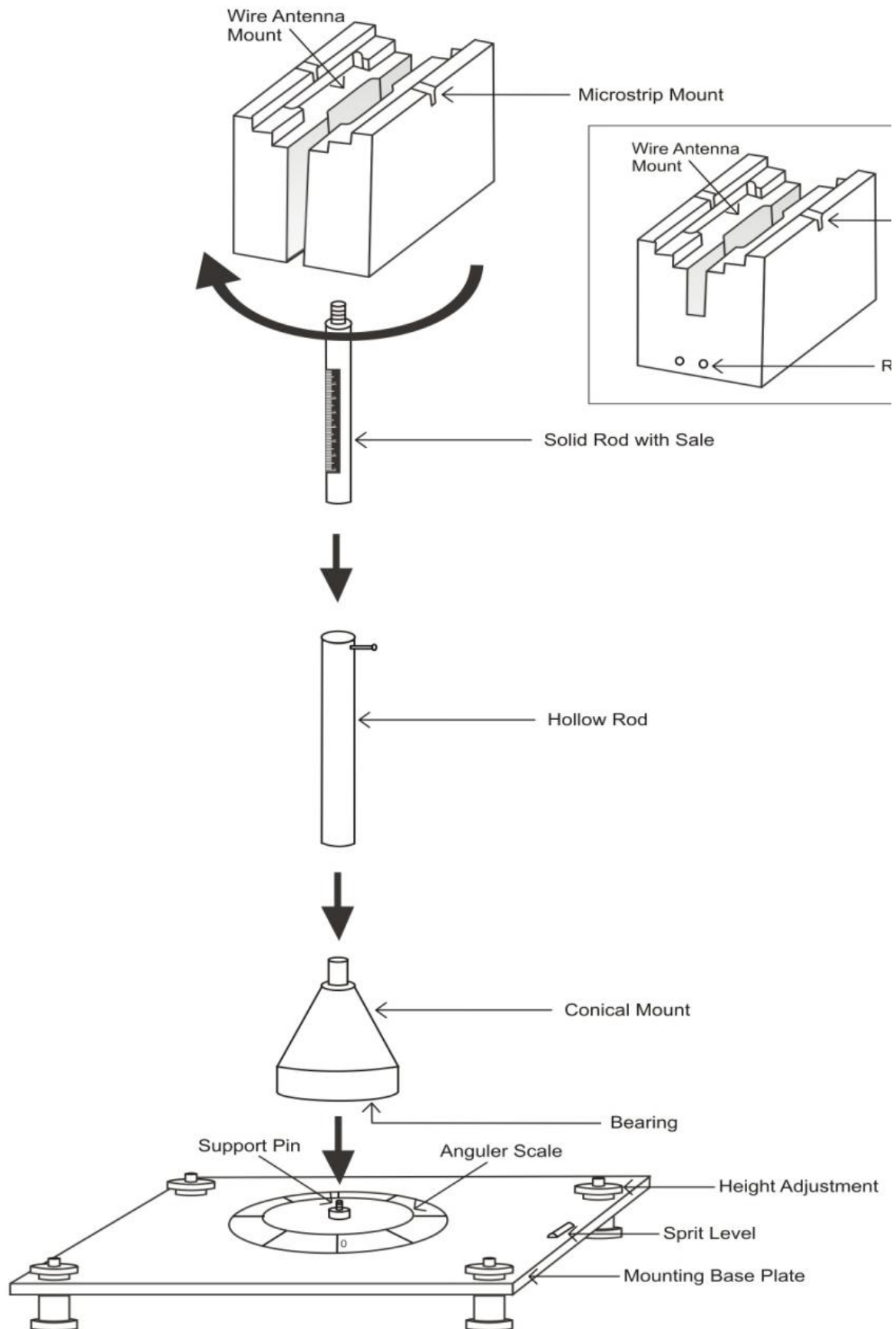


VERTICAL ASSEMBLY DESCRIPTION

1. Attach four height adjuster to base plate
2. Check the spirit level; adjust the height of the supporter until the spirit bubbles come up in the center.
3. Press fit the conical mount to base plate. To reduce the friction the bearing is present.
4. Press fit Hollow rod to conical mount
5. Fit the vertical mount to solid rod; rotate the mount clockwise so threads are fit with solid rod.
6. Insert the above assembly in hollow pipe
7. Adjust the solid rod in such a way that scale is readable while measurement, Initially do not provide height to solid rod



ASSEMBLY INSTRUCTION



TECHNICAL SPECIFICATIONS

RF Source	
Source types	PLL Synthesized based integrated with VCO
Frequency Range	100MHz to 3GHz
Frequency Resolution	100KHz
Dynamic range of transmitter	85 dBm
Transmitted power min	-4dBm (103dBuV)
Transmitted power max	+5dBm (112dBuV)
Output-frequency switching time	10ns
Tuning range	4.4GHz
Phase noise	-90dBc/Hz
Normalized Phase Noise Floor	-220Dbc/Hz
Spurious Signals Due to PFD Frequency	80 dBc
RF Detector	
Detector type	Logarithmic Detector
Frequency Range	1 MHz to 8 GHz
Resolution	0.1 dB
Dynamic range	85dB
Response time	6 ns/8 ns (fall time/rise time)
Stability over temperature	±0.5 dB (−40°C to +85°C)
Noise level	<-120dbm
Impedance	50ohm
LCD Display	20 character X 4 Line with backlit

Interface Cables	
CONNECTION	CABLE TYPE
1.Trainer board to TX antenna	RG317 SMA to SMA Male
2. Trainer board to RX antenna	RG317 SMA to SMA male
3. Antenna trainer module to PC	USB to USB cable
4. Trainer supply	Standard power supply cable 230v
Software	
	Radiation pattern plotting and analysis software suitable for windows environment
Antenna Mounting Assembly	
Antenna Positioner	• The Transmitter and Receiver Antenna stand is made of special material which is inert to EM frequency and it has engraved height and angle scale on it. It has facility to adjust the height and level. • Universal plug and fix Antenna mounts are provided to hold the antenna assembly in vertical and horizontal orientation for co and cross polarization. • Height scale for height adjustment, base plate height adjustment with spirit level.

LIST OF ANTENNA

Sr. No	Type of Antenna	Name of Antenna	Qty
1	Wire Type	Wire Dipole	2
2		Helix	1
3		Yagi-Uda 5 element	1
4		Rectangular Loop	1
5		Monopole Plane Base Ground	1
6		Folded dipole	1
7		Monopole	1

AMS-B DELIVERABLES

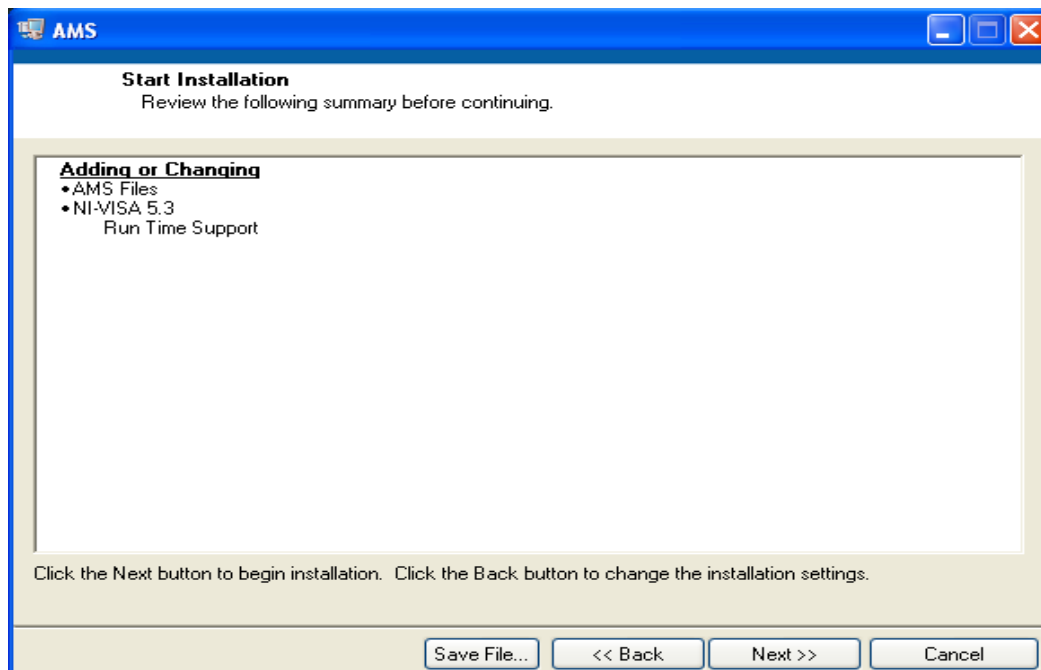
Sr. No.	Description	Quantity
1	Antenna Measurement System	1
2	SMA TO SMA MALE ADAPTER	1
3	ANTENNA MOUNTING ASSAMBLY	
	A. Height Adjuster Legs	8
	B. Base plate for Transmitter	1
	C. Base plate for Receiver with Conical Mount	1
	D. Conical mount for Transmitter	1
	E. Hollow rod	2
	F. Solid rod with scale	2
	G. Big Nut for Solid rod of Transmitter	1
	H. Universal Mount with Screws	1
	I. Screw Driver	1
	J. Allen Key	1

Accessories:

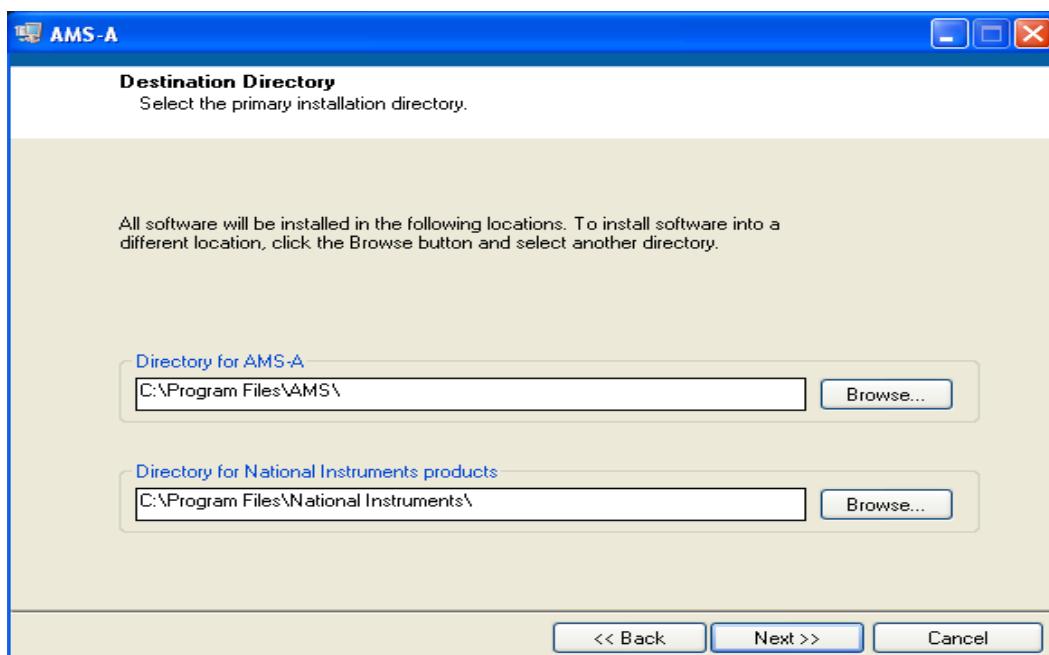
	RFL-AMS-B	Quantity
1	USB A male To USB A male cable	1
2	SMA male to SMA male 2 mtr cable	1
3	SMA male to SMA male 1 mtr cable	1
4	RFL-AMS-B Expt Manual	1
5	Software CD	1
6	Power cord	1

AMS-B SOFTWARE INSTALLATION

1. Run the setup.exe available from the AMS installer folder (*CD provided with kit*), then following window will appear, click on Next.



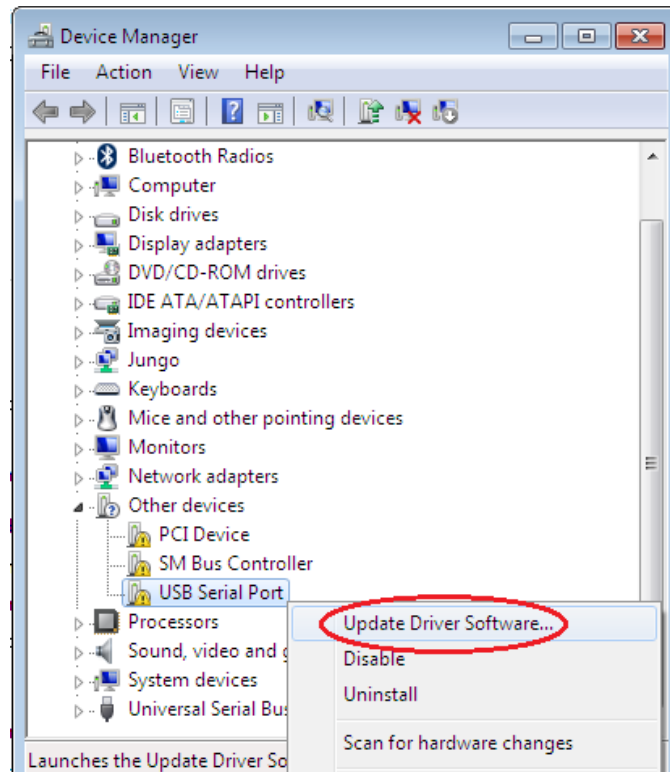
2. Select the directory where you want to install the setup, and then click on Next. Then click on finish



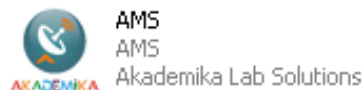
3. Restart the PC.
4. Connect the **AMS-B (TRAINER KIT)** to the **PC** using USB to USB cable (*provided with the kit*). Sometime FTDI Drivers get installed automatically or sometime we you have to install them.

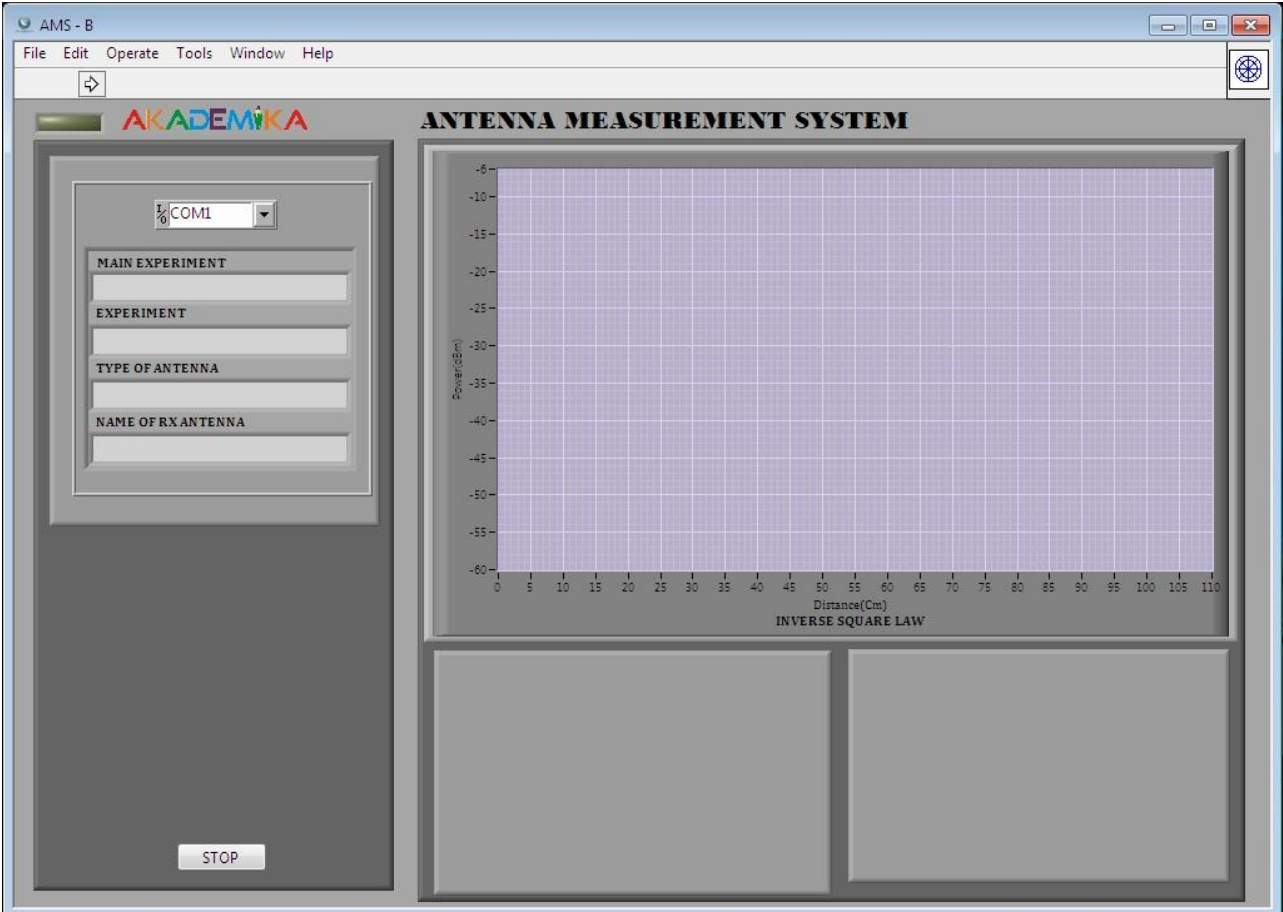
5. For installation of FTDI Drivers manually follow the procedure:

- Right click on **My Computer**
- **Properties**
- **Device Manager**
- **Others**
- Right Click on **USB Serial Port** & select **Update Driver Software**



- **Browse my Computer for Driver Software**
 - Select proper folder for Driver software from CD provided with kit
 - **Click Next**
 - Follow the same procedure once again then you will get **USB Serial Port COM** Address.
6. After the installation, go to directory where software installation is taken place and open the **AMS-B**





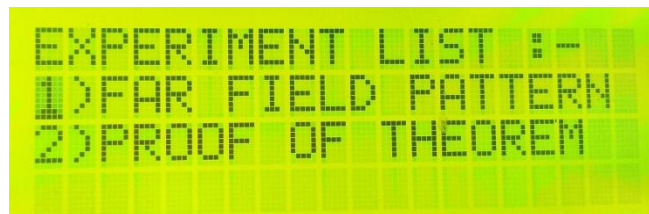
PANEL CONTROL MODE

1. When we turn on the kit following message will be displayed.



PRESS MENU KEY TO
SELECT EXPERIMENT

2. Select Main Experiment as **Far Field Pattern**.



EXPERIMENT LIST :-
1) FAR FIELD PATTERN
2) PROOF OF THEOREM

3. Select **Co-Polarization** as Experiment from the panel.



FAR FIELD PATTERN
1) CO-POLARIZATION
2) CROSS-POLARIZATION
3) GAIN/DIRECTIVITY

4. Use **SHIFT** button to Scroll. Select Type of Antenna as **WIRE**.



CO-POLARIZATION
1) REFLECTOR
2) APERTURE
3) WIRE

5. Select Antenna as **CIRCULAR LOOP**.



WIRE
1) BOOTAM FEED LOOP
2) CROSS DIPOLE
3) CIRCULAR LOOP

6. Now select the experiment.

```

CIRCULAR LOOP
>EXPERIMENT
2>VIEW READINGS

```

7. Make the dipole length as displayed on the screen.

```

MOUNT TX&RX ANTENNA
MAKE SURE THAT
DIPOLE LENGTH=16.5cm
CONFIRM????

```

8. Set Receiver Antenna to 0^0 and Press ENTER key to start the experiment.

```

SET RX ANTENNA TO 0
DEGREE AND PRESS
ENTER KEY TO START
EXPERIMENT

```

9. Now the selected antenna will be displayed.

```

X = DIPOLE
X = CIRCULAR LOOP
DIPOLE LENGTH=16.5cm
TX FREQ = 1000MHz

```

10. Rotate Rx antenna by 5° & simultaneously press ENTER key.

```

RECEIVED POWER
ANGLE POWER(dBm)
10 -38.2182

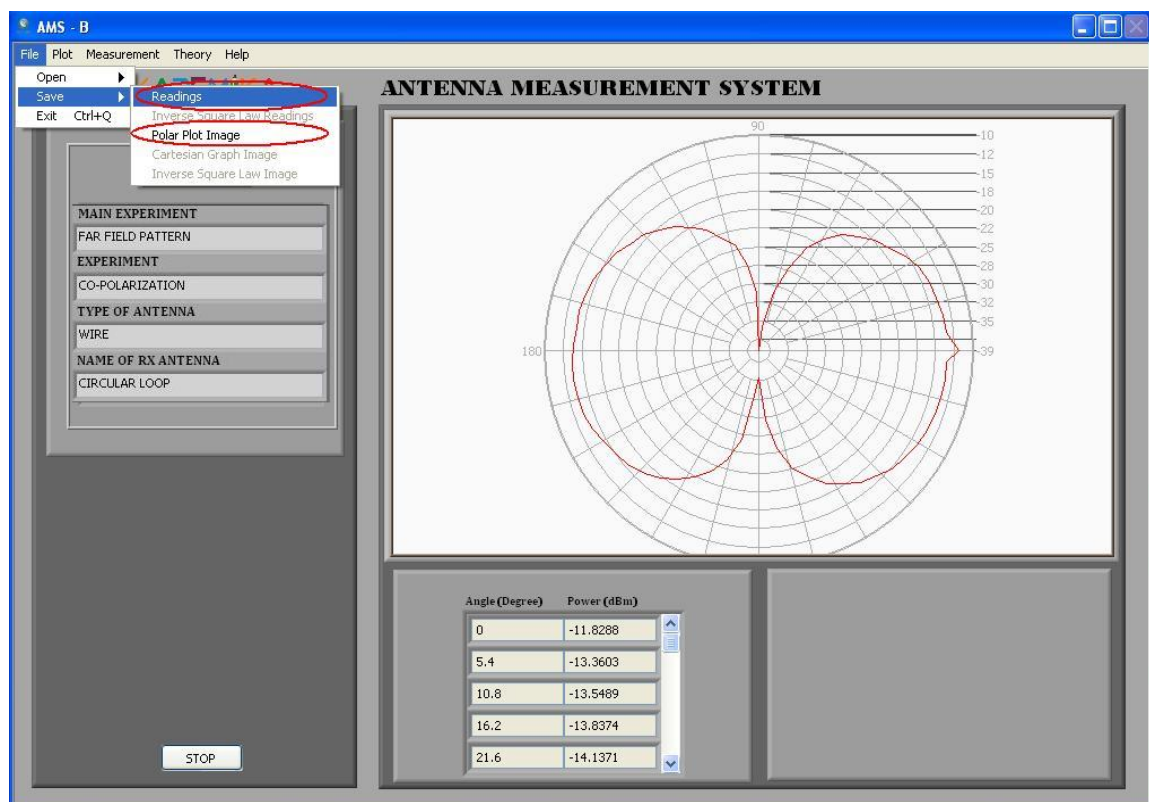
```

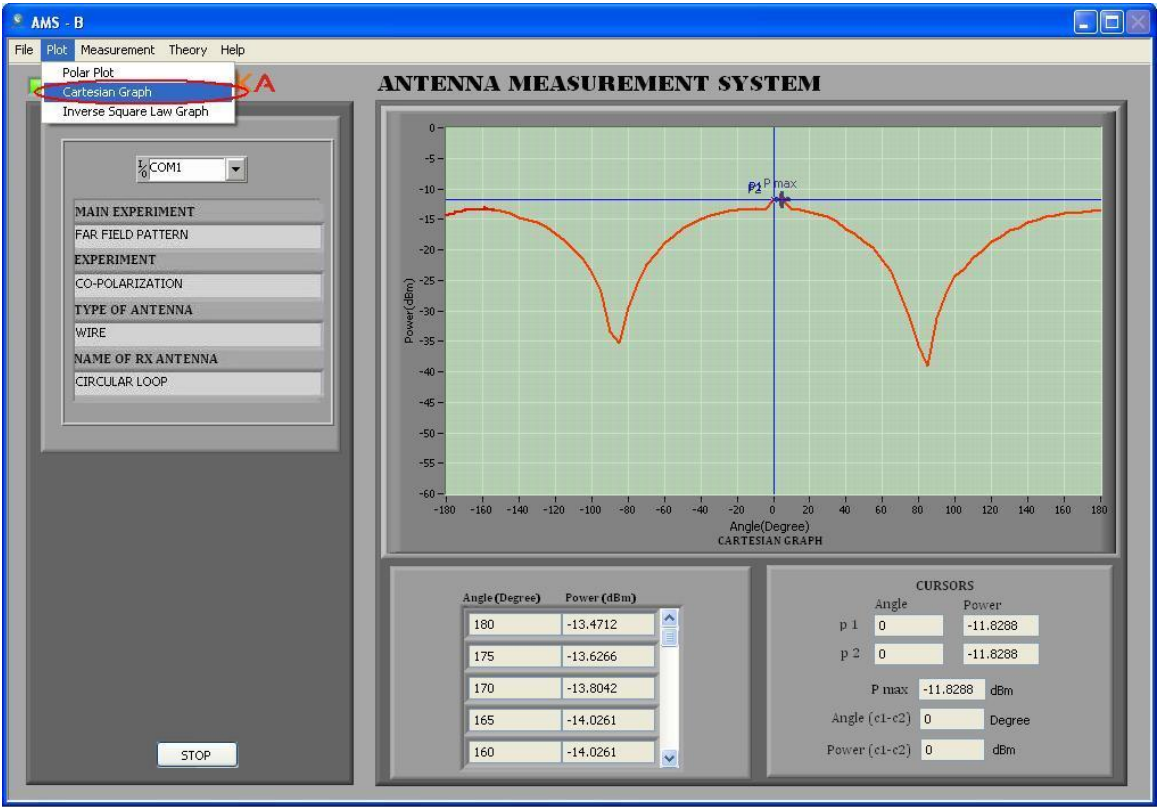

11. After completion of all the 68 readings, it will ask to upload the Readings on Computer. Confirm the selection using ENTER button. Make sure the software is Running and press ENTER again.

SAVE AND UPLOAD DATA
TO COMPUTER??????

MAKE SURE THAT SOFT-
WARE MUST BE RUNNING

12. It will then plot the graph on Software screen PULLING UP the Experiment details and Antenna selected, as shown in fig.





EXPERIMENT NO. 1

EXPERIMENT NO: 1

TO PROVE INVERSE SQUARE LAW

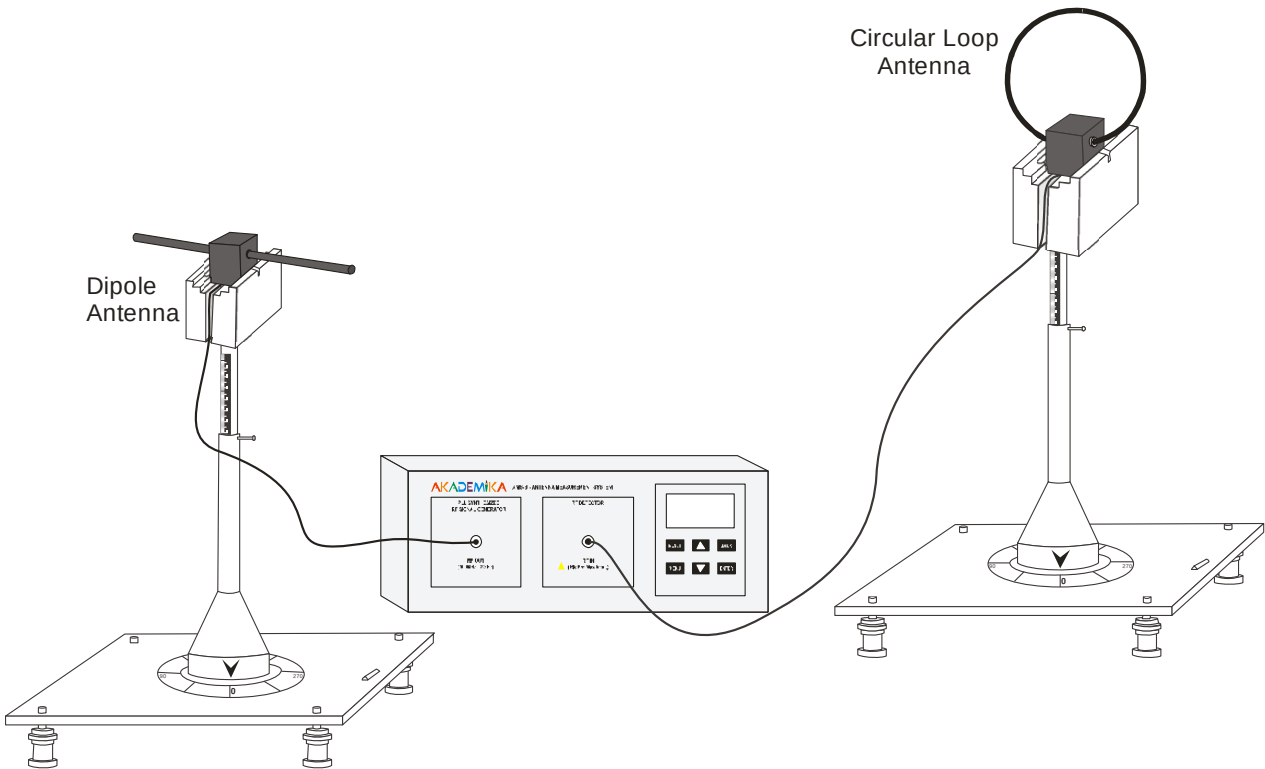
OBJECTIVE

To Measure the variation of field strength of radiated wave, with distance from transmitting antenna.

EQUIPMENT REQUIRED:

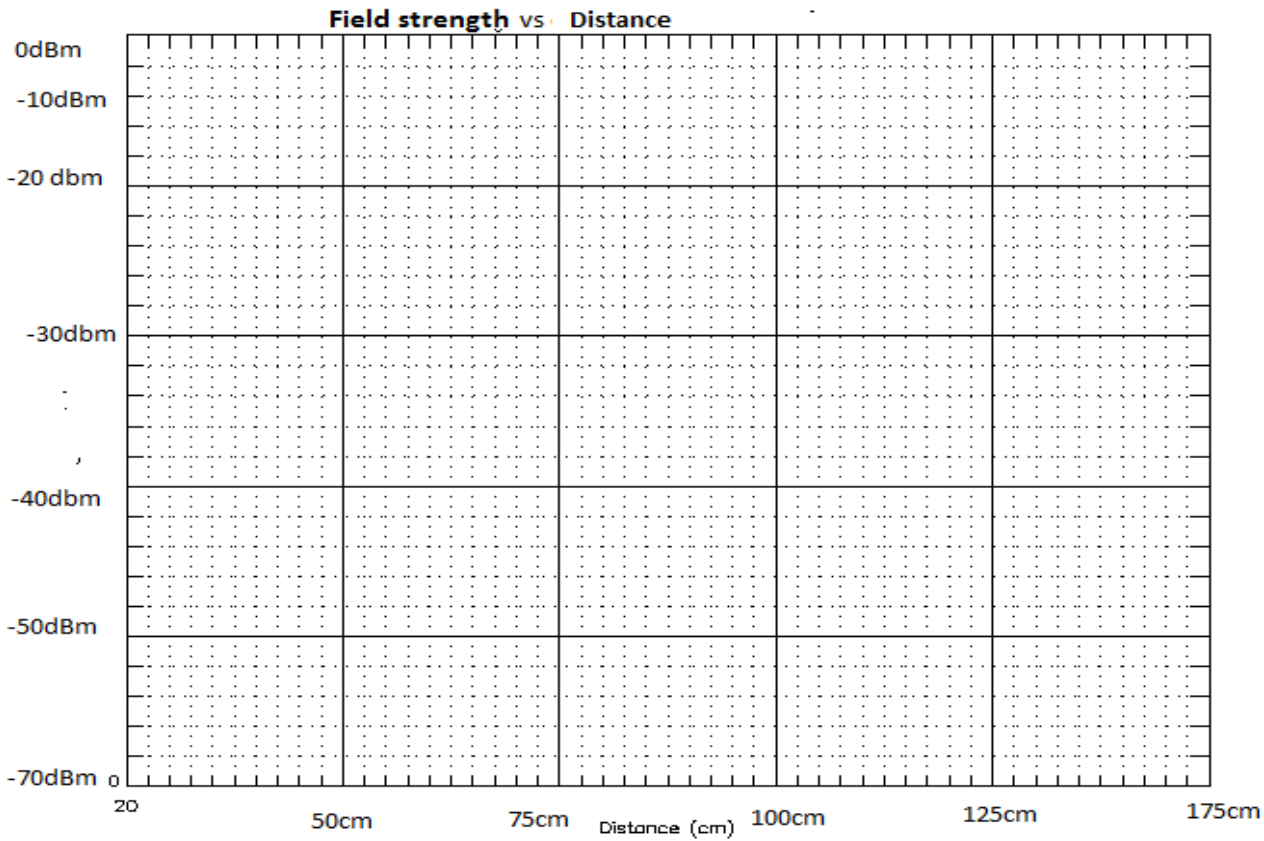
TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire or Microstrip DIPOLE	Any antenna	Source (RF output)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SETUP ARRANGEMENT:



PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Turn ON the module.
3. Open the AMS-B.exe file Select the corresponding COM PORT and Click on
4. Run, now the software will be in running mode.
5. Select the PROOF OF THEOREM and then select the experiment of
6. INVERSE SQUARE LAW.
7. Then select the RX antenna and click on START button.
8. A Pop-Up window will appear asking user to Keep the Distance between TX and RX as 10cm then click on NEXT button.
9. After completion of one reading again a popup window will appear asking to
10. Increase the TX and RX distance by 10cm.
11. Repeat the procedure up to a distance of 100cm at an interval of 10cm; click on NEXT to plot the graph.
12. After that it will plot the radiation pattern in inverse square law graph (i.e.
13. Distance in cm V/S power in dBm).
14. While taking the readings, ensure that no scattering objects are in the vicinity
15. of the antenna, this could reradiate and distort the field pattern and consequently the readings. Avoid any movement of persons while taking the readings.
16. Plot these readings manually on graph paper with distance between antennas on X axis and signal level in dB at Y-axis.
17. Use the graph template provided below for plotting your graph.
18. Now take the readings and observe the plots for different types of antenna.



EXPERIMENT NO. 2

EXPERIMENT NO. 2

TO PROVE RECIPROCITY THEOREM

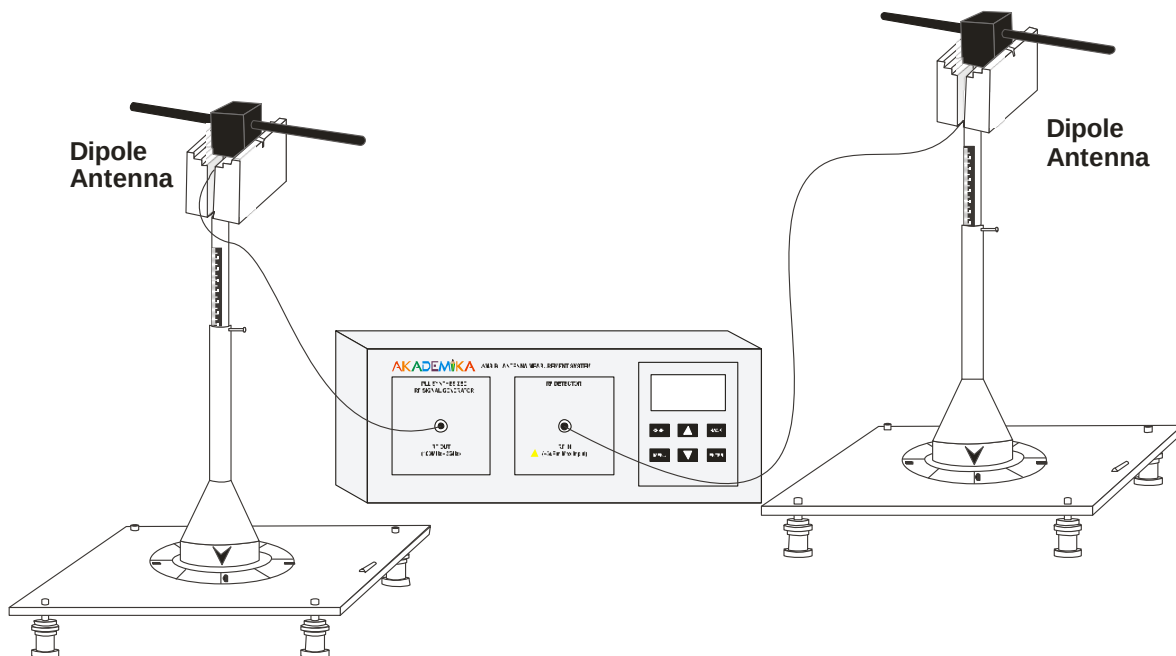
OBJECTIVE

To demonstrate that the transmitting and receiving radiation patterns of an antenna are equal and hence confirm the reciprocity theorem of antennas.

EQUIPMENT REQUIRED:

TX antenna	RX antenna	Transmitter input	Receiver output	Cable
1) Dipole 2) RMSA	1) Dipole 2) RMSA	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SETUP ARRANGEMENT:



PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 1meter, consult theory
3. for details of far field region.
4. Turn ON the module.
5. Open the AMS-B.exe file Select the corresponding COM PORT and Click on
6. Run, Now the software will be in running mode.
7. Select the PROOF OF THEOREM and then select the experiment of RECIPROCITY THEOREM.
8. Then select the RX antenna and click on START button.
9. After completing it will plot the readings.
10. Now interchange the antenna and again take the next reading.
11. Compare both the plots they must be same, hence reciprocity theorem is proved.
12. Similarly do the experiments for different types of antenna.

NOTE:

1. **Keep in mind that an antenna that is being rotated is plotted in reception and transmission mode both for proving the reciprocity theorem.**
2. **Observe the two plots and they must be approximately same.**

EXPERIMENT NO. 3

EXPERIMENT NO. 3

RADIATION PATTERN OF WIRED ANTENNA

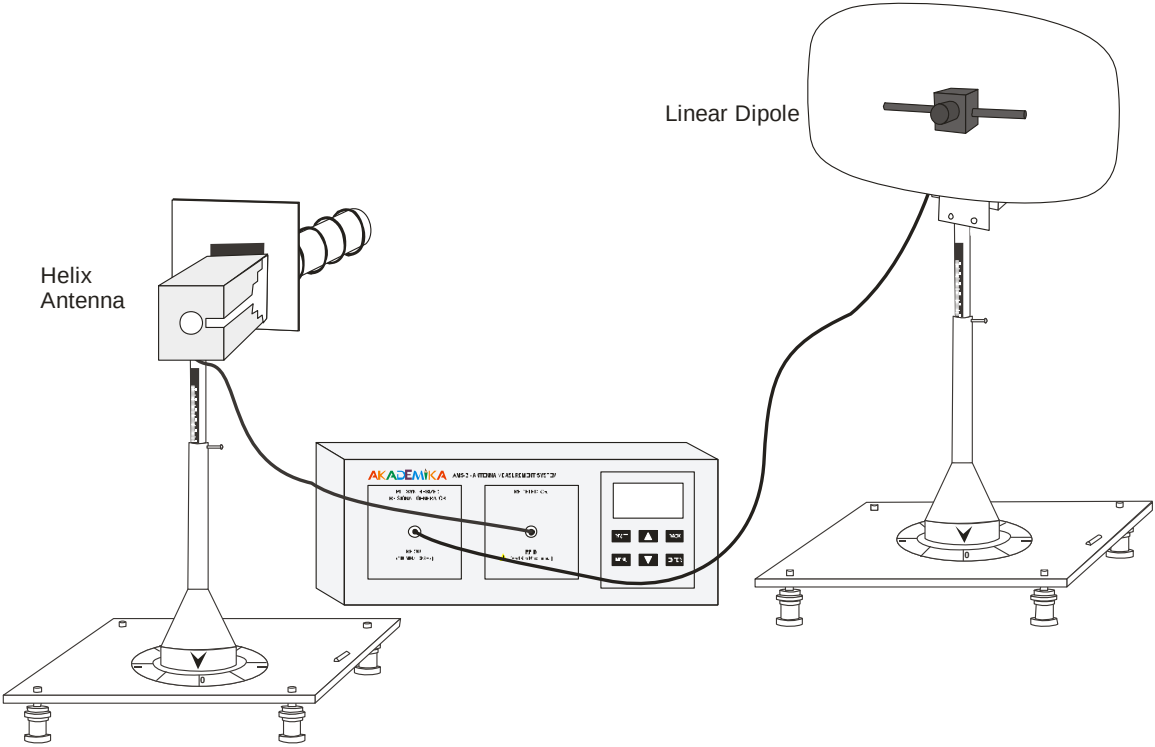
OBJECTIVE

To plot the Radiation pattern of WIRED antenna and observe its parameters.

EQUIPMENT REQUIRED:

TX Antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire DIPOLE	Any wired antenna	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SET UP ARRANGEMENT:

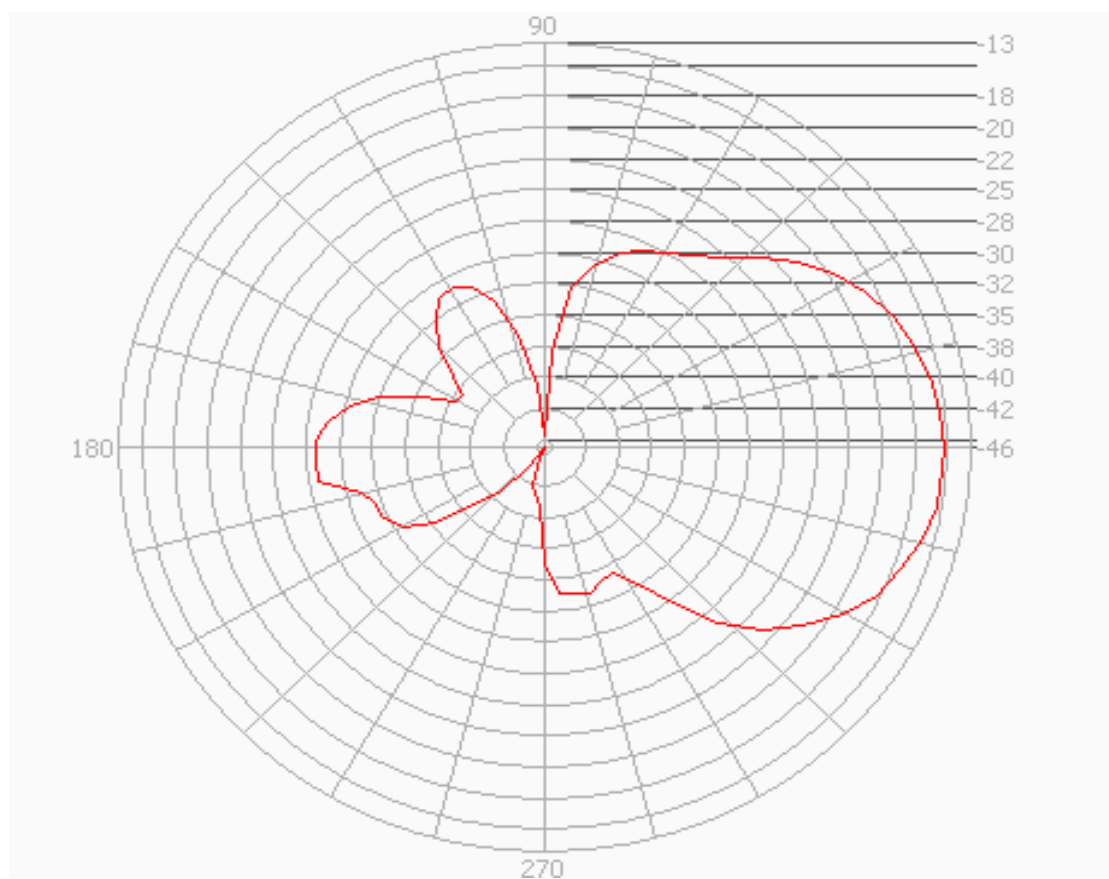


PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 1meter, consult theory for details.
3. Turn ON the module.
4. Open the AMS-B.exe file; select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in FAR FIELD PATTERN, select CO-POLARIZATION, select RX antenna then click on START
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. Now repeat for CROSS-POLARIZATION of antenna and observe the plot.

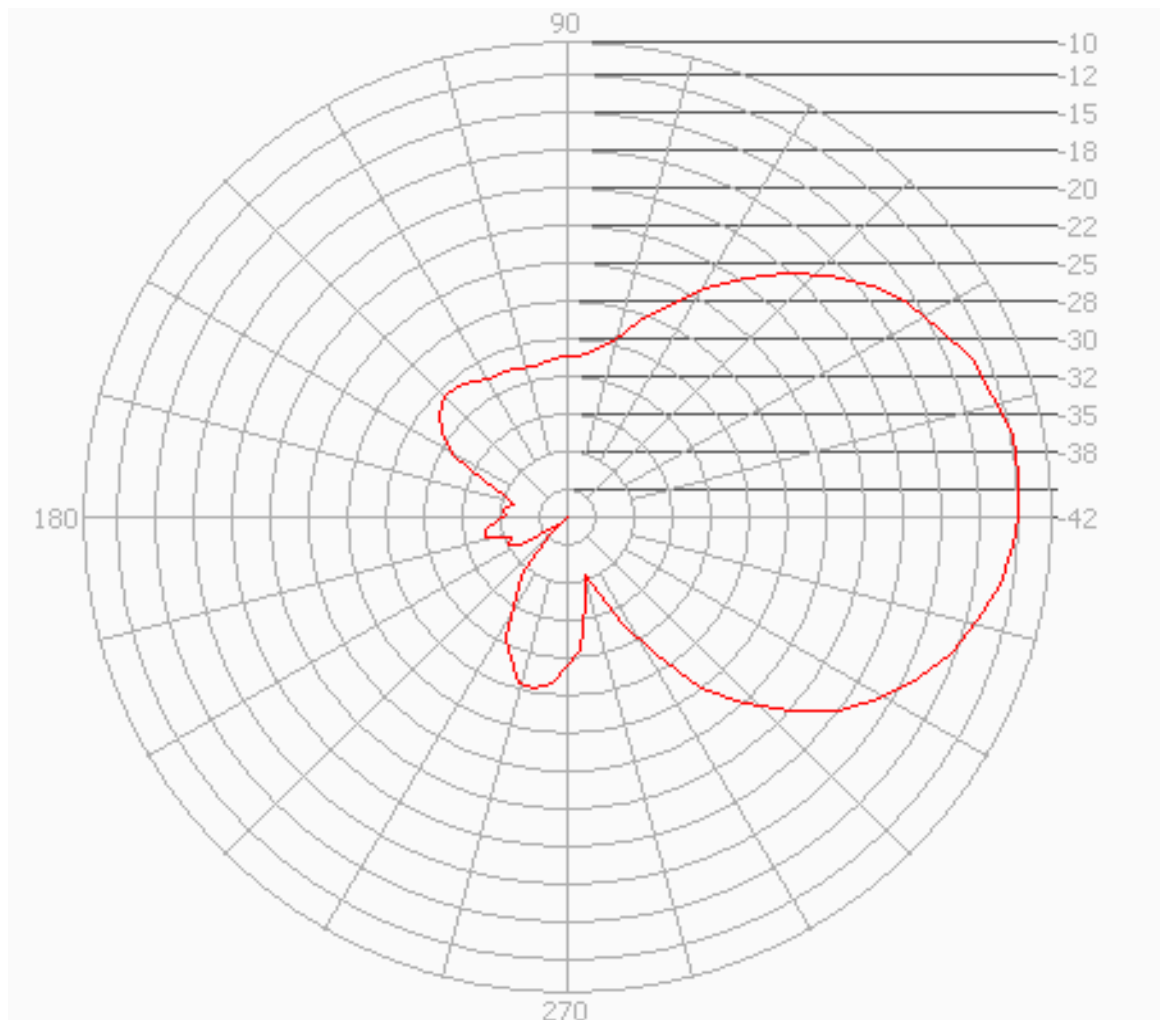
Example 1

TX antenna	RX antenna	Transmitter input	Receiver output	Cable
Wire Dipole	Helix	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN:

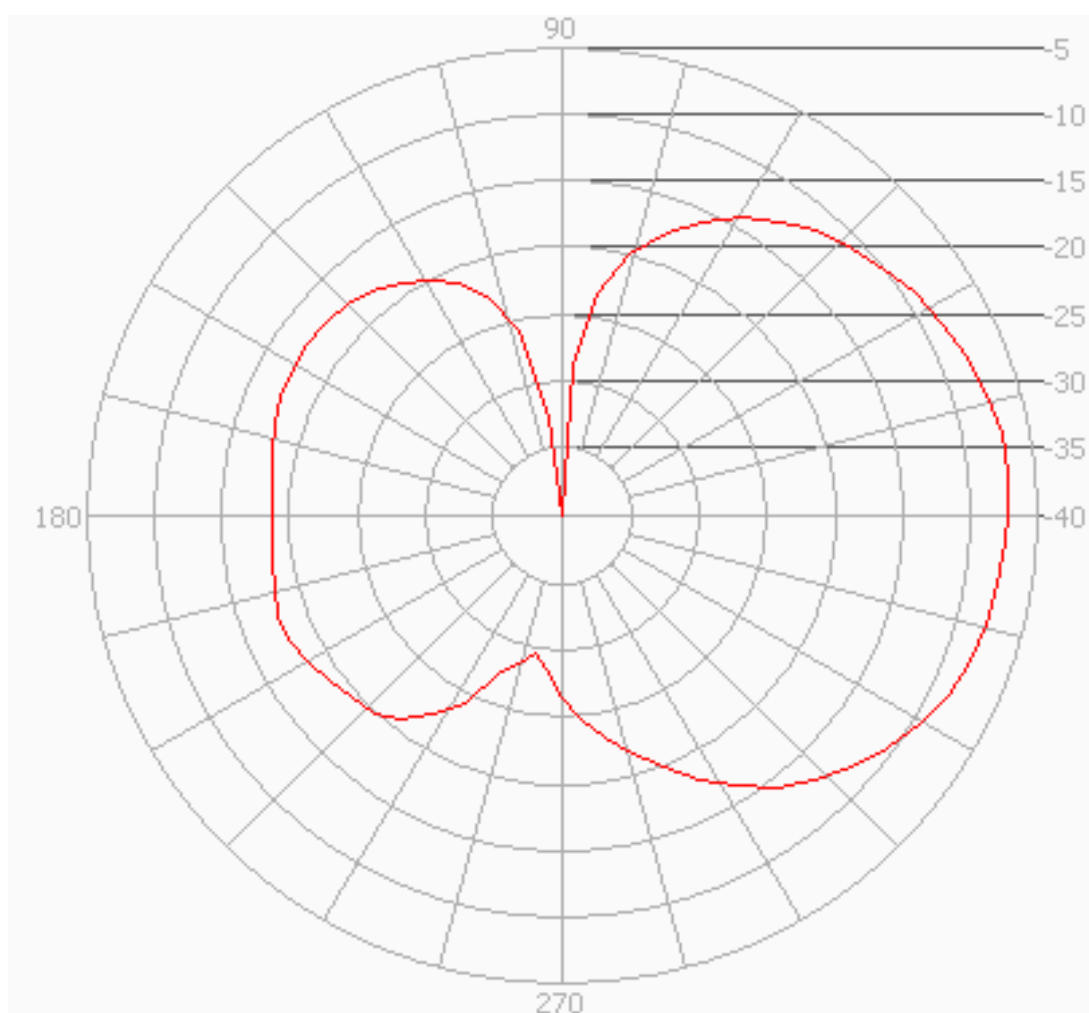
EXAMPLE 2

TX antenna	RX antenna	Transmitter input	Receiver output	Cable
Wire Dipole	Log periodic	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN:

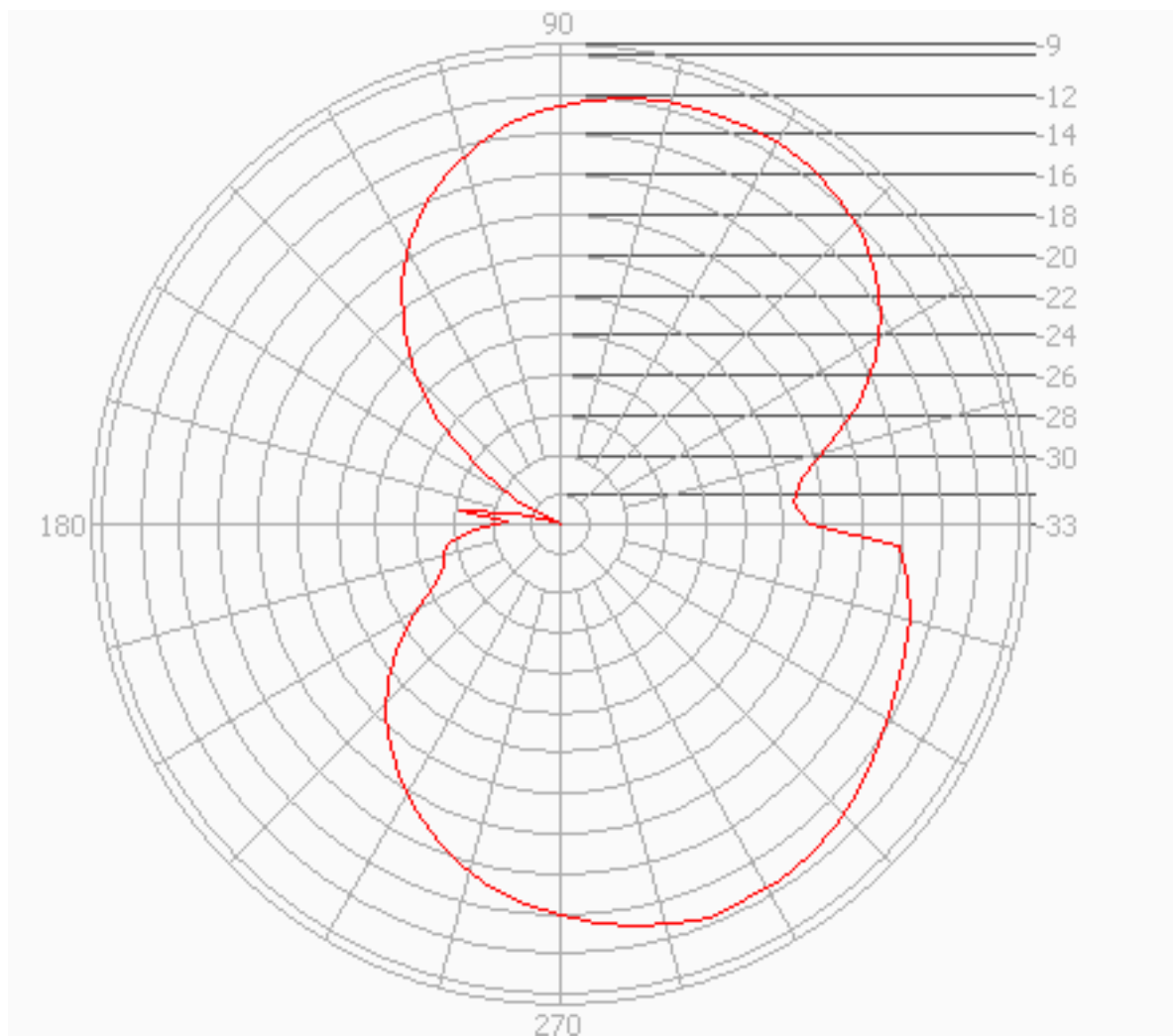
EXAMPLE 3

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire Dipole	Yagi UDA (5 elements)	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN:

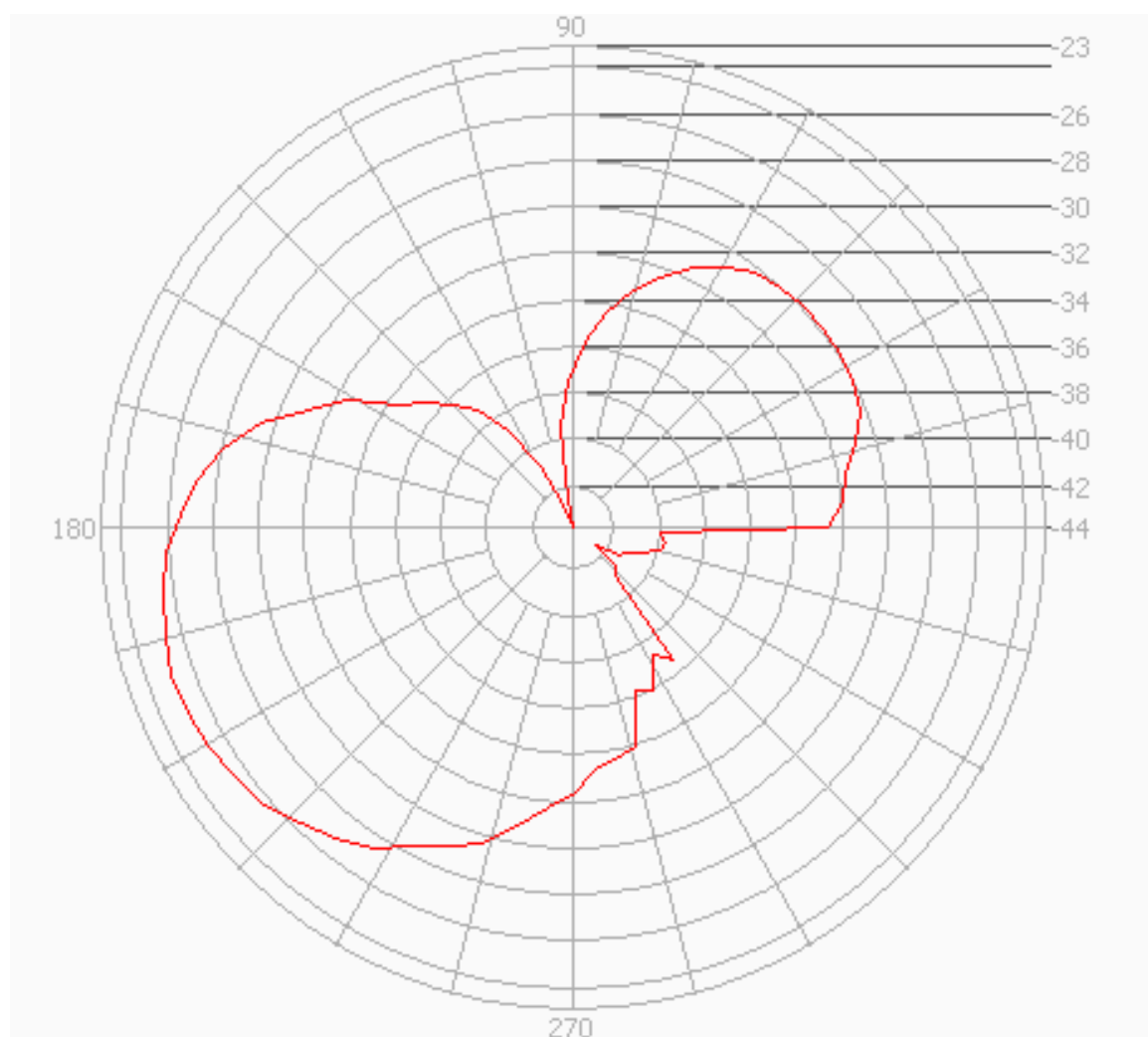
EXAMPLE 4

TX antenna	RX antenna	Transmitter input	Receiver output	Cable
Wire Dipole	Wire monopole	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN:

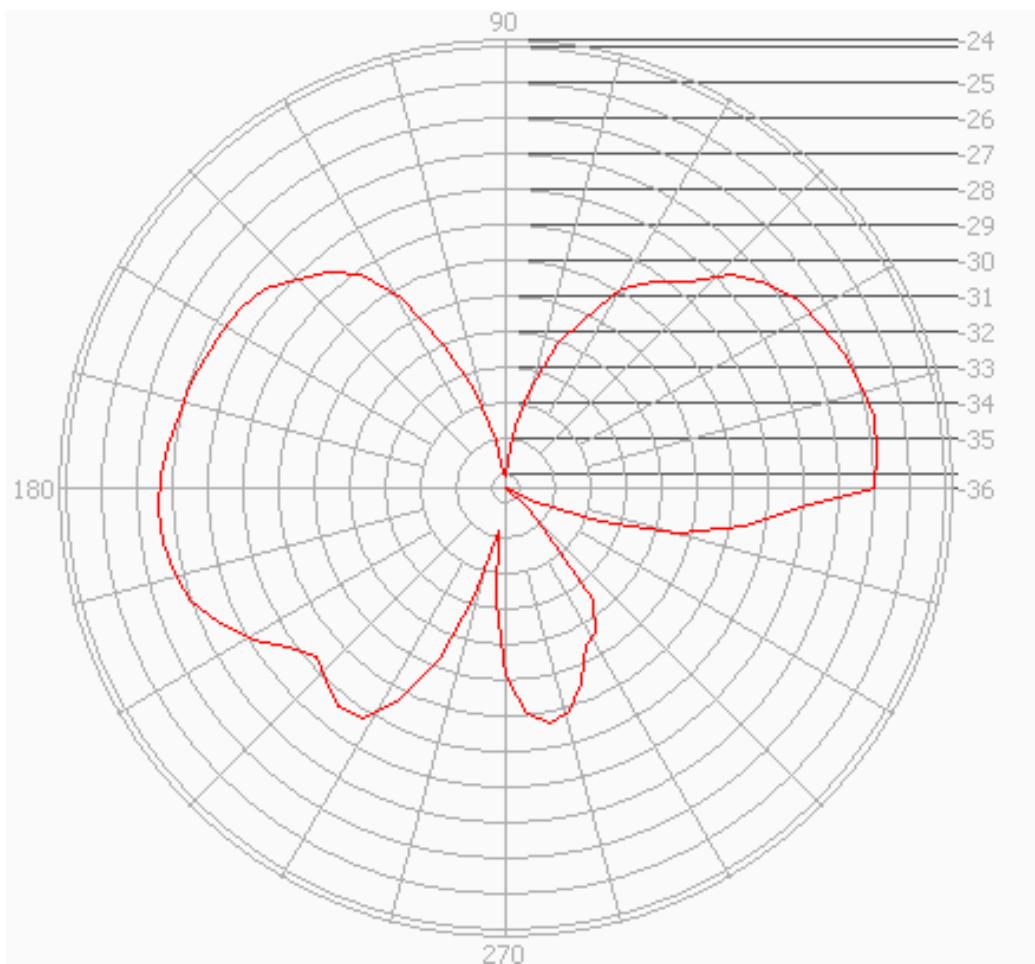
EXAMPLE 5

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire Dipole	Rectangular loop	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN:

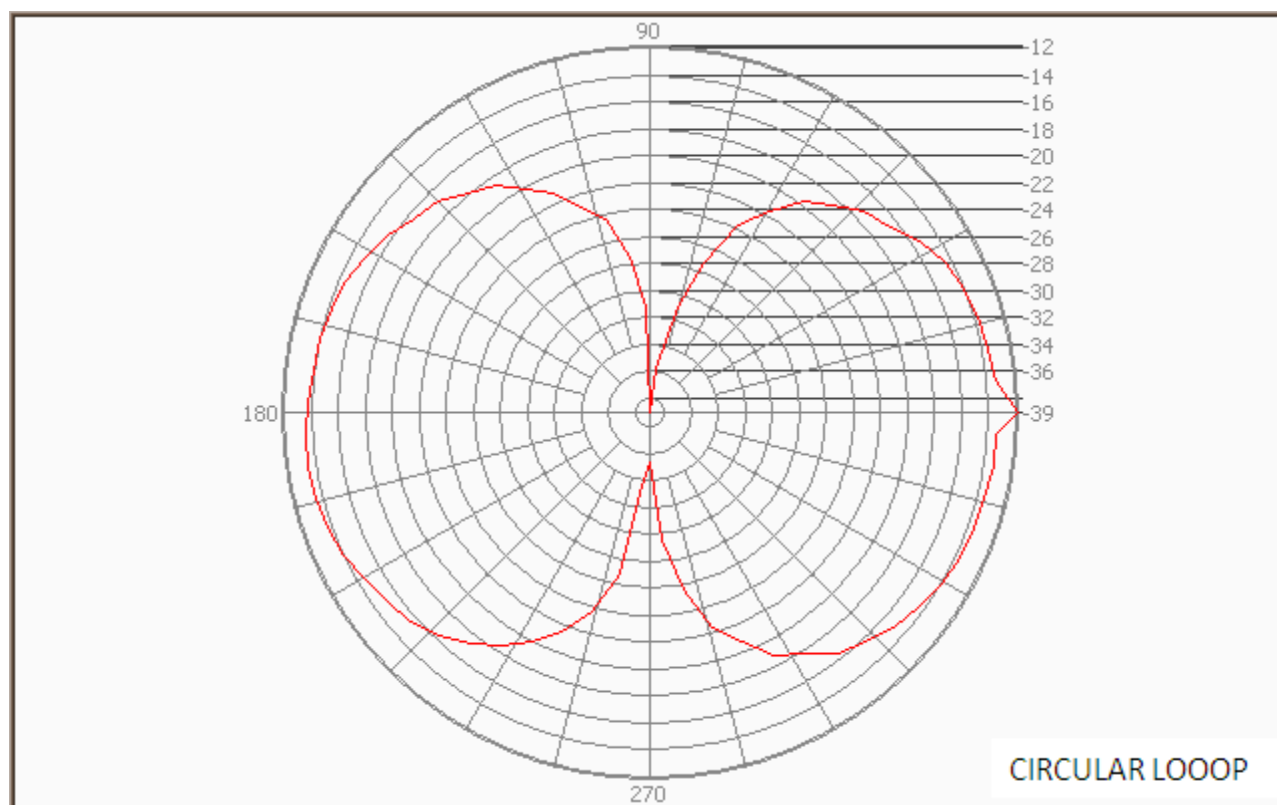
EXAMPLE 6

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire Dipole	VEE	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN

EXAMPLE 7

TX antenna	RX antenna	Transmitter input	Receiver output	Cable
Wire Dipole	Circular loop	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN:

**EXPERIMENT
NO. 4**

EXPERIMENT NO. 4

RADIATION PATTERN OF APERTURE ANTENNA

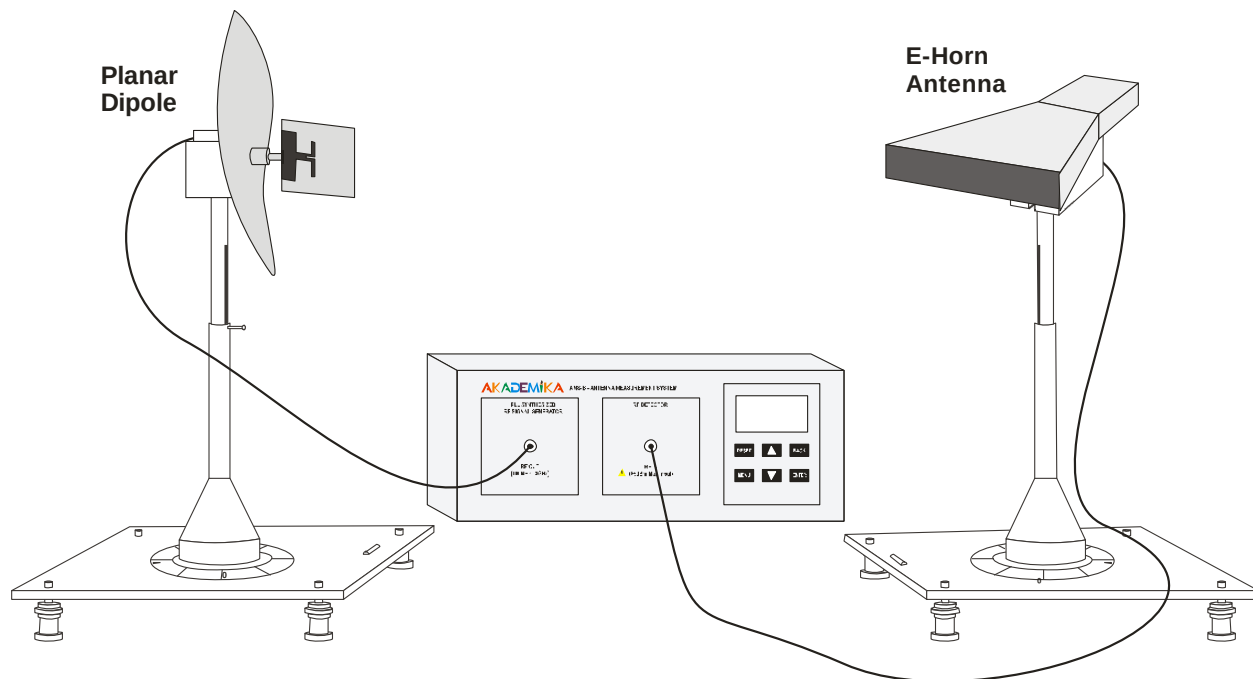
OBJECTIVE

To plot the radiation pattern of all APERTURE antenna and observe its parameters.

EQUIPMENT REQUIRED:

TX Antenna	RX antenna	Transmitter input	Receiver output	Cable
Planar DIPOLE	Any APERTURE antenna	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SET UP ARRANGEMENT:



PROCEDURE:

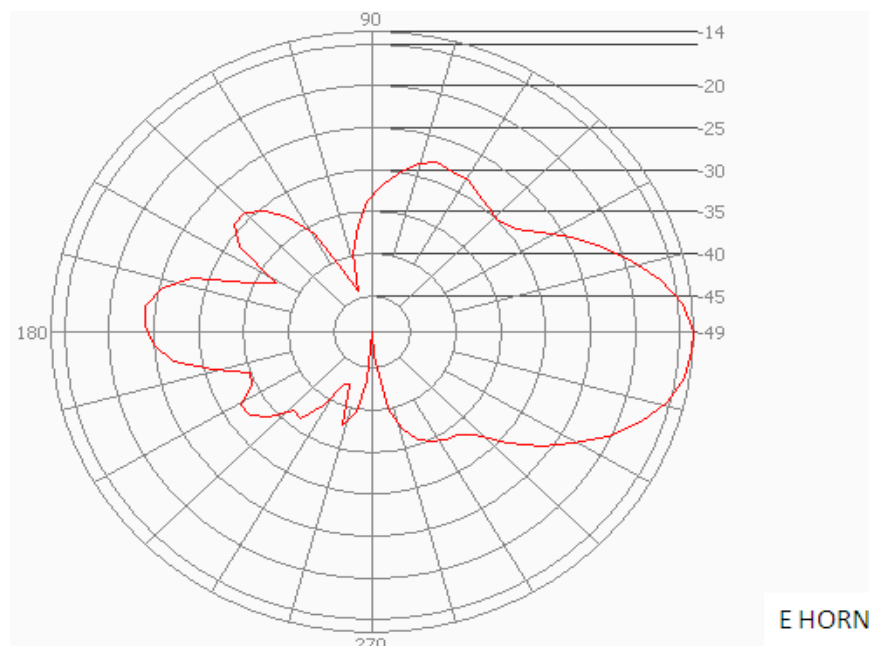
1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 1meter, consult theory for details.
3. Turn ON the module.

4. Open the AMS-B.exe file; select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in FAR FIELD PATTERN, select CO-POLARIZATION, select RX antenna then click on START
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. Now repeat for CROSS-POLARIZATION of antenna and observe the plot.

EXAMPLE

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Planner Dipole	E-Horn	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN



EXPERIMENT NO. 5

EXPERIMENT NO. 5

RADIATION PATTERN OF REFLECTOR ANTENNA

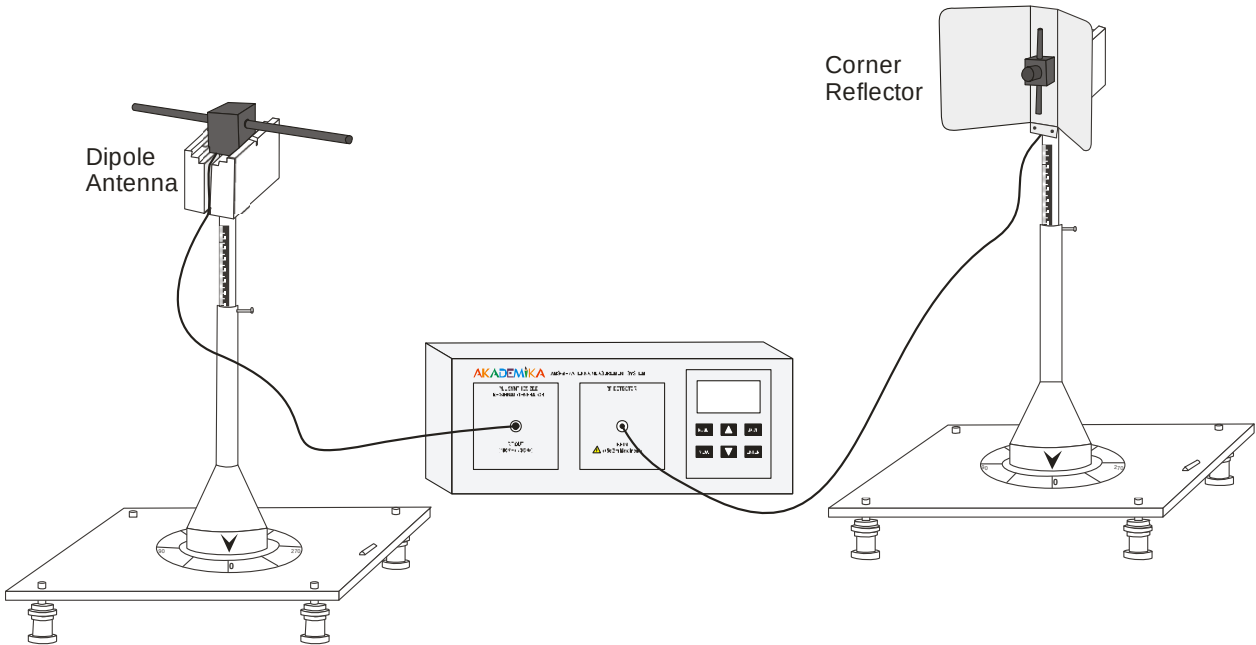
OBJECTIVE

To plot the radiation pattern of REFLECTOR antenna and observe its parameters.

EQUIPMENT REQUIRED:

TX Antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire DIPOLE	Any REFLECTOR antenna	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SETUP ARRANGEMENT:

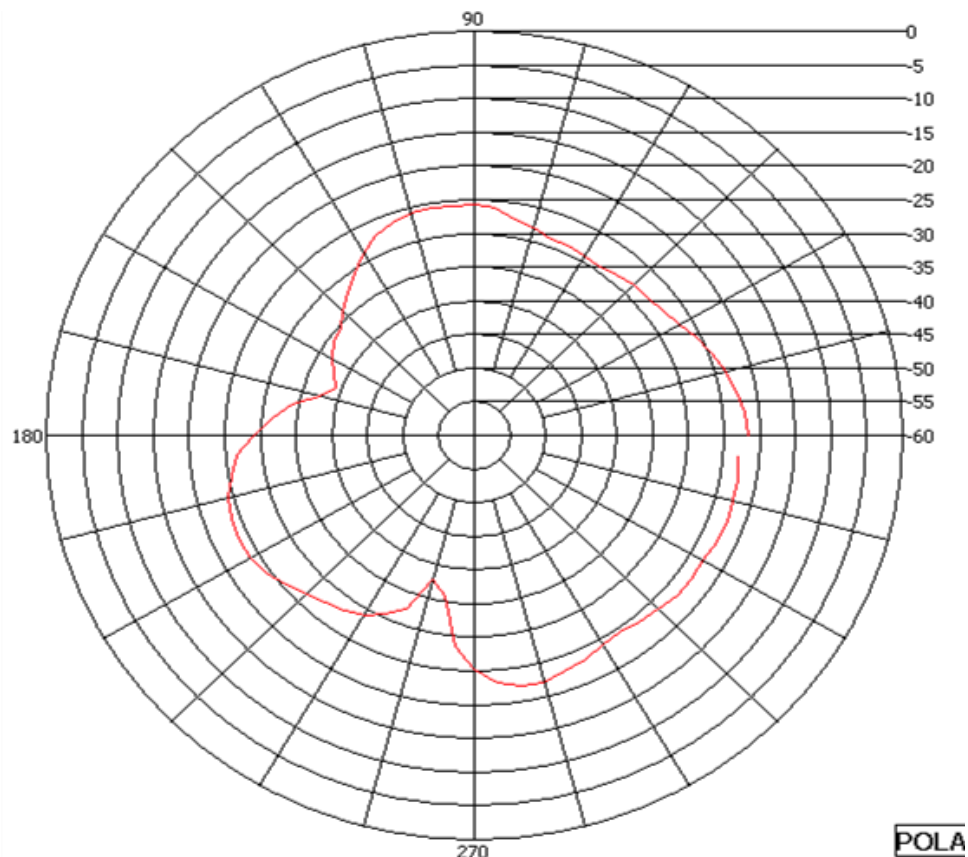


PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 1meter, consult theory for details.
3. Turn ON the module.
4. Open the AMS-B.exe file; select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in FAR FIELD PATTERN, select CO-POLARIZATION, select RX antenna then click on START
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. Now repeat for CROSS-POLARIZATION of antenna and observe the plot.

EXAMPLE

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire Dipole	Corner reflector	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN:

**EXPERIMENT
NO. 6**

EXPERIMENT NO. 6

RADIATION PATTERN OF ARRAY ANTENNA

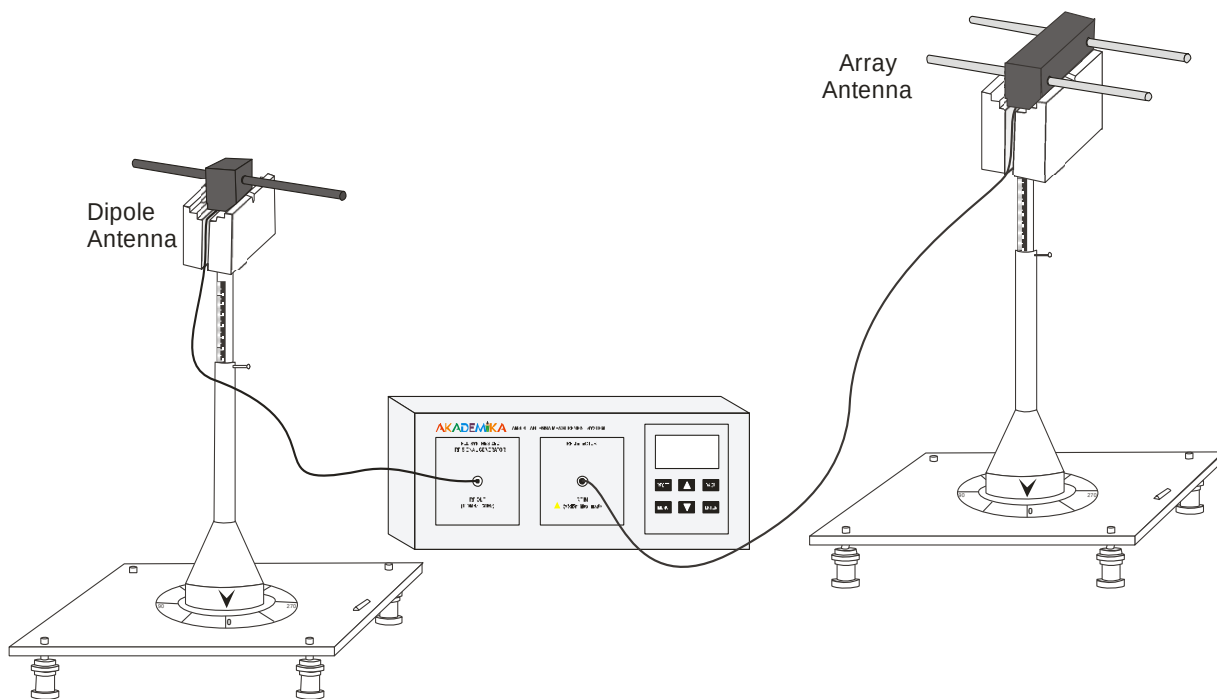
OBJECTIVE

To plot the radiation pattern of ARRAY antenna and observe its parameters.

EQUIPMENT REQUIRED:

TX Antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire DIPOLE	Any ARRAY antenna	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SET UP ARRANGEMENT:

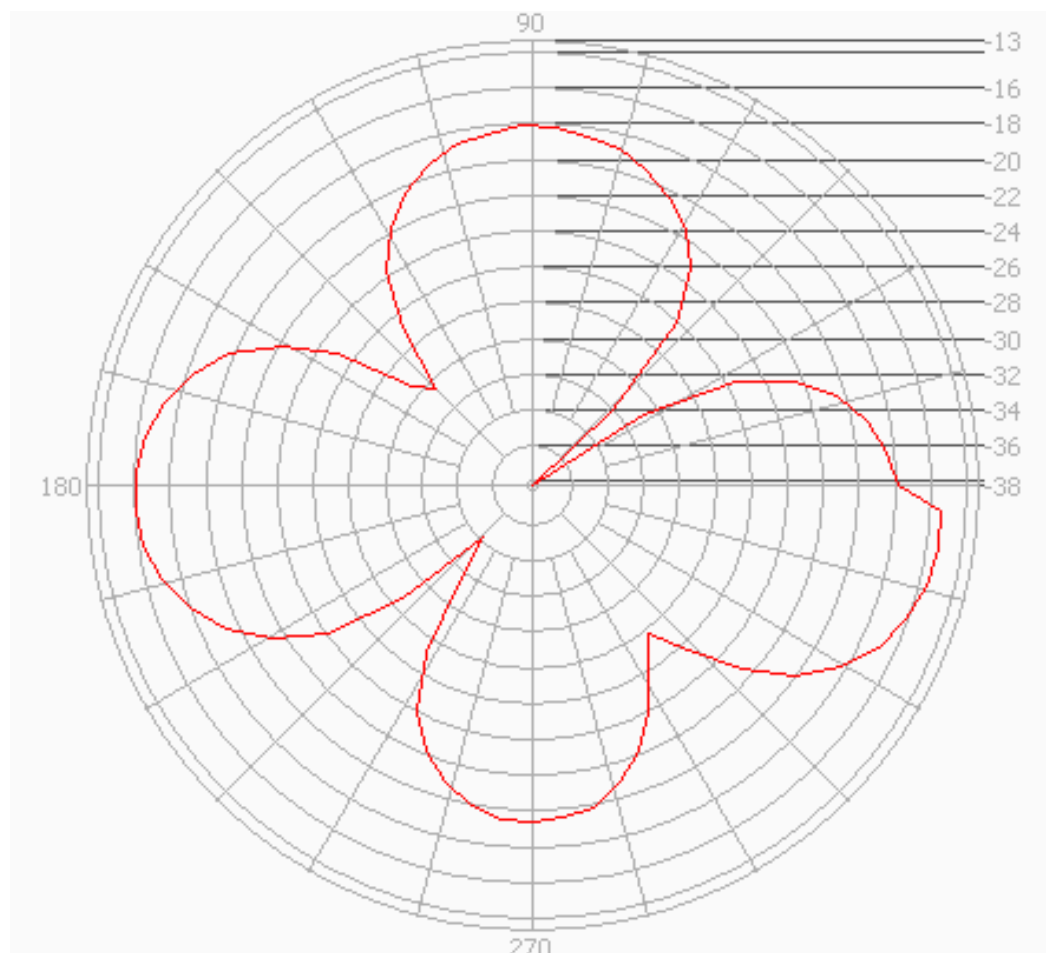


PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 1meter, consult theory for details.
3. Turn ON the module.
4. Open the AMS-B.exe file; select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in FAR FIELD PATTERN, select CO-POLARIZATION, select RX antenna then click on START
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. Now repeat for CROSS-POLARIZATION of antenna and observe the plot.

EXAMPLE 1

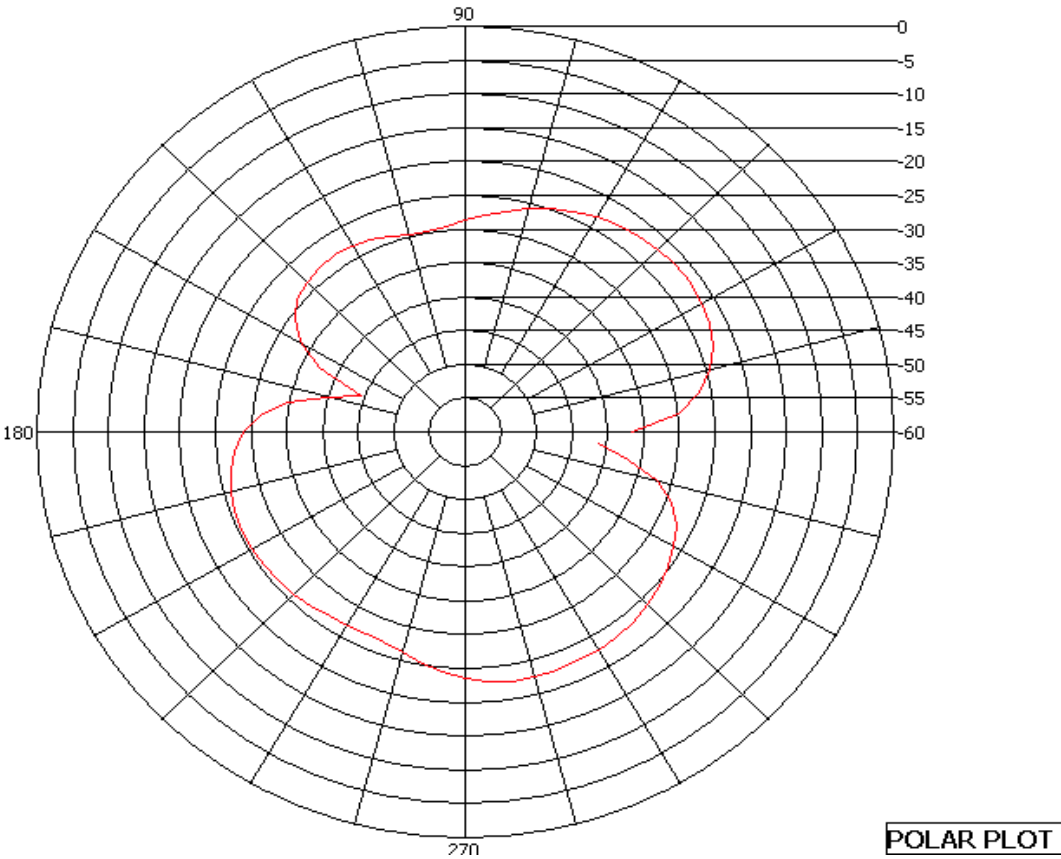
TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire Dipole	Broadside array	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN:

EXAMPLE 2

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire Dipole	End fire	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN:



**EXPERIMENT
NO. 7**

EXPERIMENT NO. 7

CO-POLARIZATION AND CROSS-POLARIZATION

OBJECTIVE

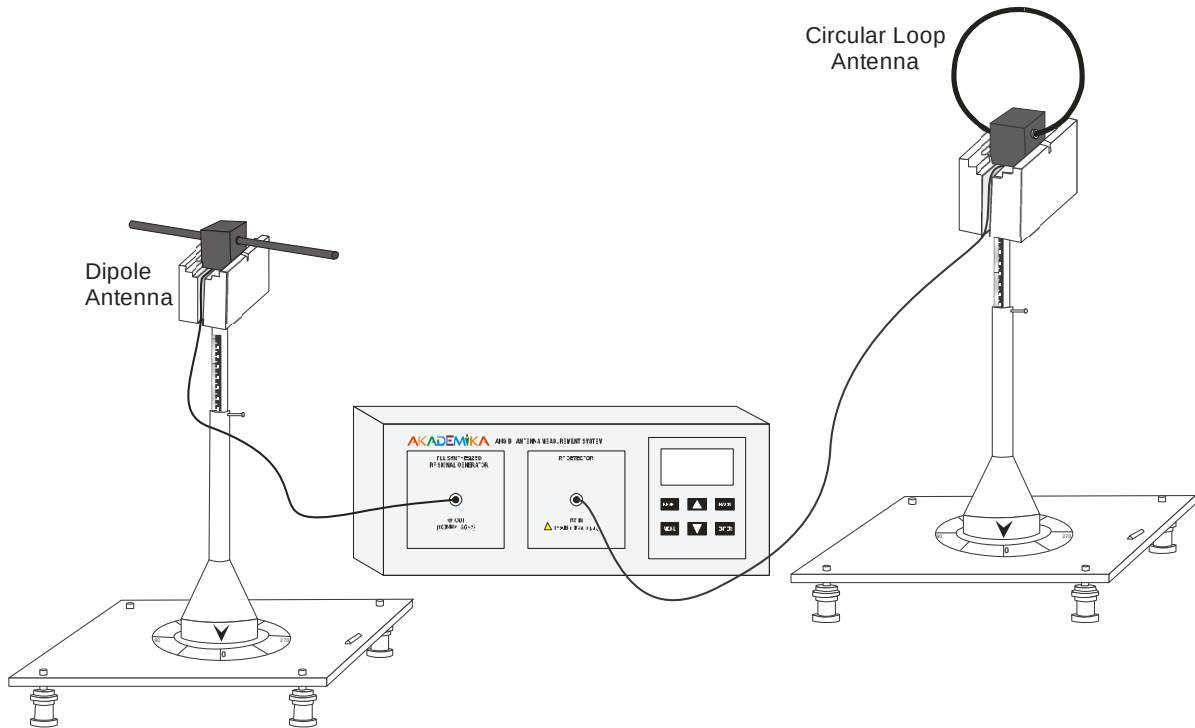
To observe the effect of CO-POLARIZATION and CROSS-POLARIZATION using vertical and horizontal polarization of antenna.

EQUIPMENT REQUIRED:

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
1. WIRE DIPOLE 2. PLANAR(Microstrip) DIPOLE	1. Any WIRE antenna 2. Any MICROSTRIP antenna	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

PROCEDURE:

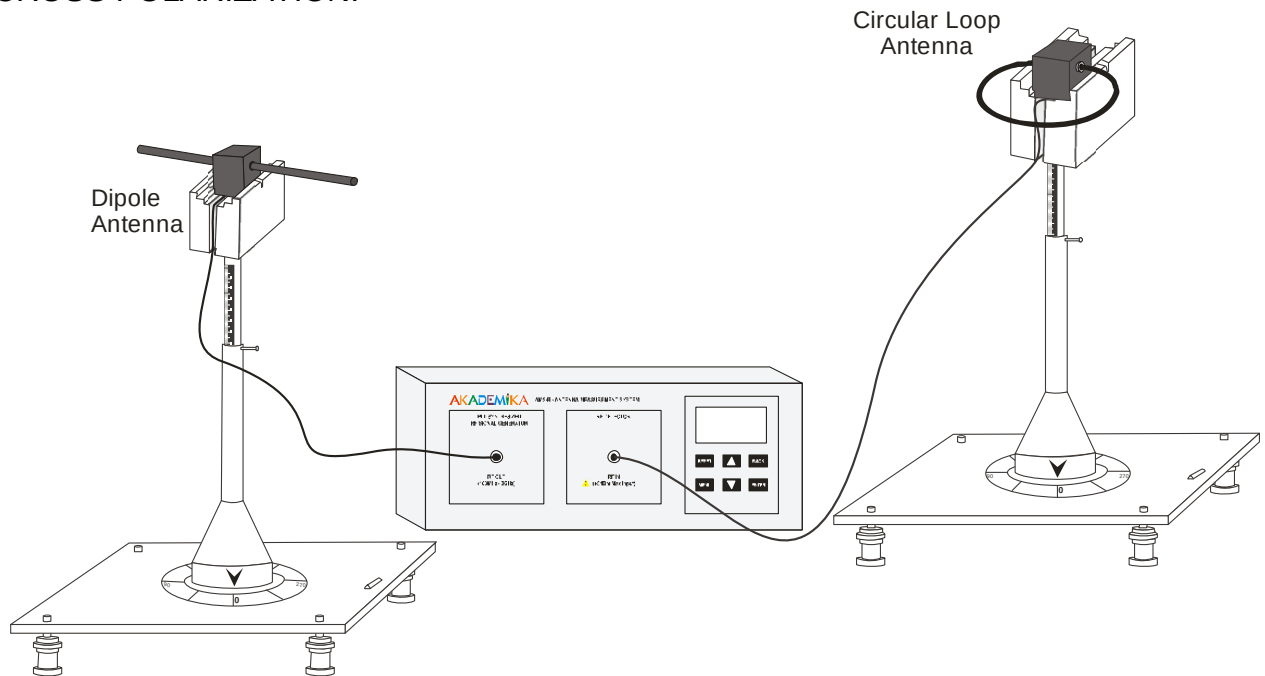
CO-POLARIZATION:



1. Set up the experiment as per shown in figure above.
2. Set transmitting as well as receiving antenna in same polarization i.e. VERTICAL/HORIZONTAL.

3. Set the distance between the antennas to be around 1meter, consult theory for details of far field region.
4. Turn ON the module.
5. Open the AMS-B.exe file; select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
6. Go in FAR FIELD PATTERN, select CO-POLARIZATION, select RX (either WIRE or PLANAR) antenna then click on START
7. Then readings from 0 degree to 360 degree will be plotted in the software.
8. After completing it will plot the readings.

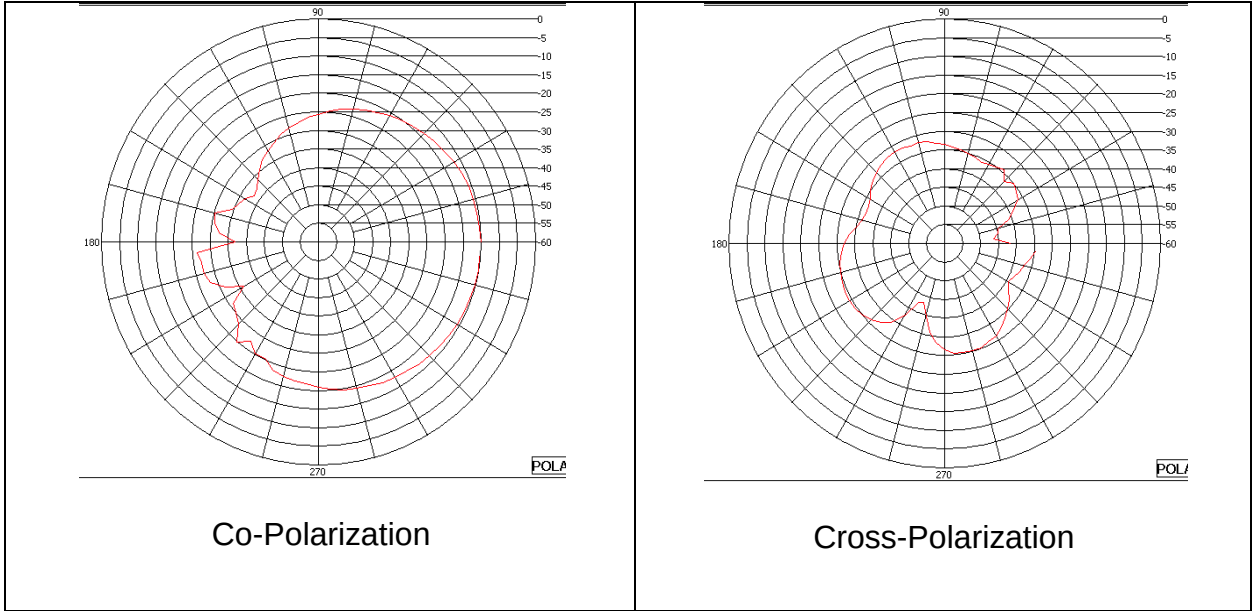
CROSS POLARIZATION:



1. Set up the experiment as per shown in figure above.
2. Set transmitting as well as receiving antenna in opposite/different polarization i.e. VERTICAL/HORIZONTAL.
3. Set the distance between the antennas to be around 1meter, consult theory for details of far field region.
4. Open the AMS-B.exe file; select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in FAR FIELD PATTERN, select CROSS-POLARIZATION, select RX (either WIRE or PLANAR) antenna then click on START.
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. After completing it will plot the readings.
8. Observe the effect of change in polarization of antenna.

EXAMPLE:

TX antenna	RX antenna	TX frequency	Separation	Measurement
Planar Dipole	RMSA	2428MHz	60cm	Co polarization Cross-Polarization



**EXPERIMENT
NO. 8**

EXPERIMENT NO. 8

CIRCULARLY POLARIZED ANTENNAS

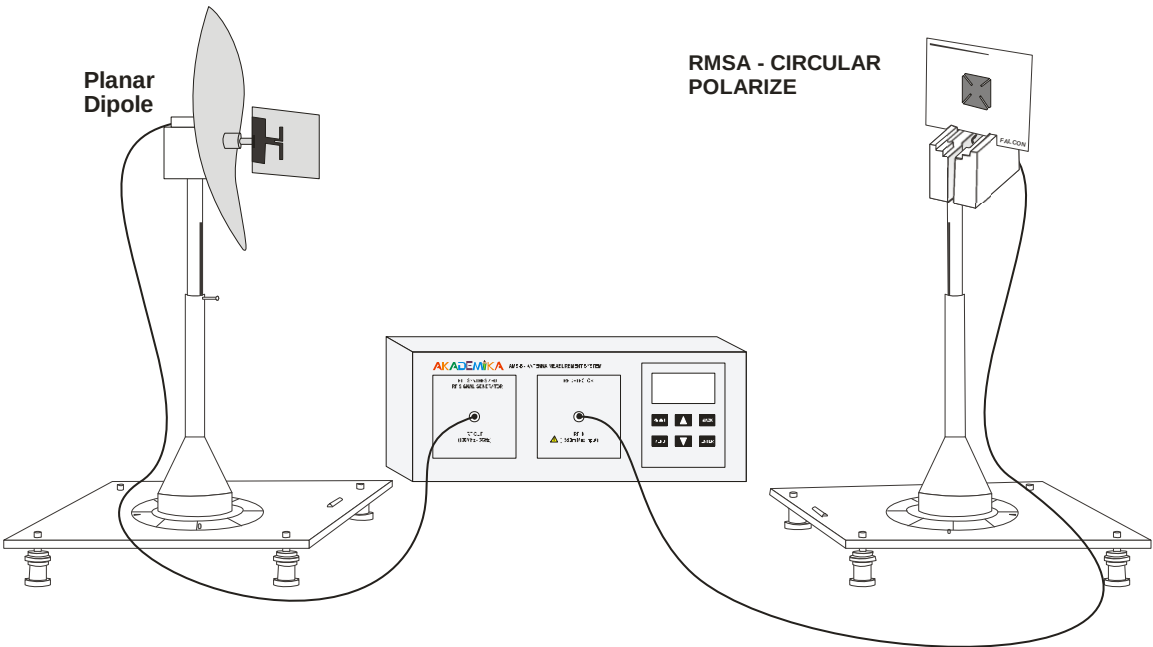
OBJECTIVE

Measurement of Circularly polarized antennas and observe its parameters.

EQUIPMENT REQUIRED:

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
1. WIRE DIPOLE 2. PLANAR (Microstrip) DIPOLE	1. Helix 2. Cross Dipole 3. RMSA-Circular Polarized	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SETUP ARRANGEMENT:

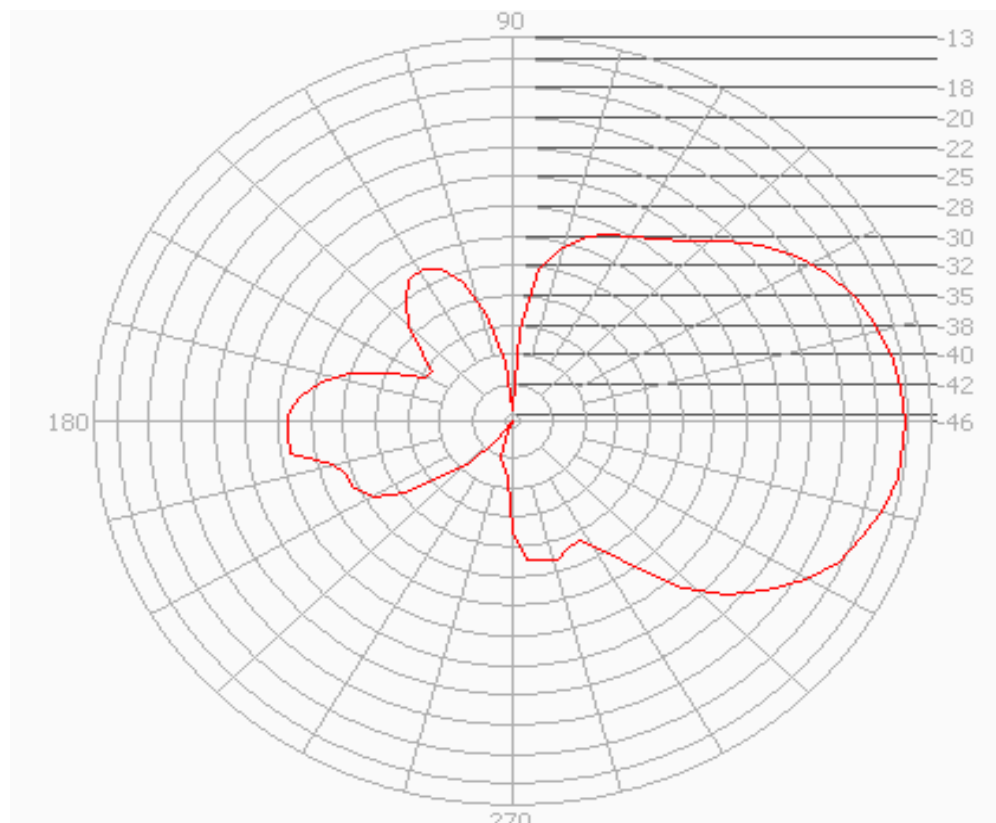


PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 1meter, consult theory for details.
3. Turn ON the module.
4. Open the AMS-B.exe file; select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in FAR FIELD PATTERN, select CO-POLARIZATION, select RX antenna then click on START
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. Now repeat for CROSS-POLARIZATION of antenna and observe the plot.
8. Thus the circular polarized antenna receives a maximum power in both the polarization.

EXAMPLE 1

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire Dipole	Helix	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN:

**EXPERIMENT
NO. 9**

EXPERIMENT NO. 9.1

RADIATION PATTERN OF ALL PLANAR (MICROSTRIP) ANTENNA

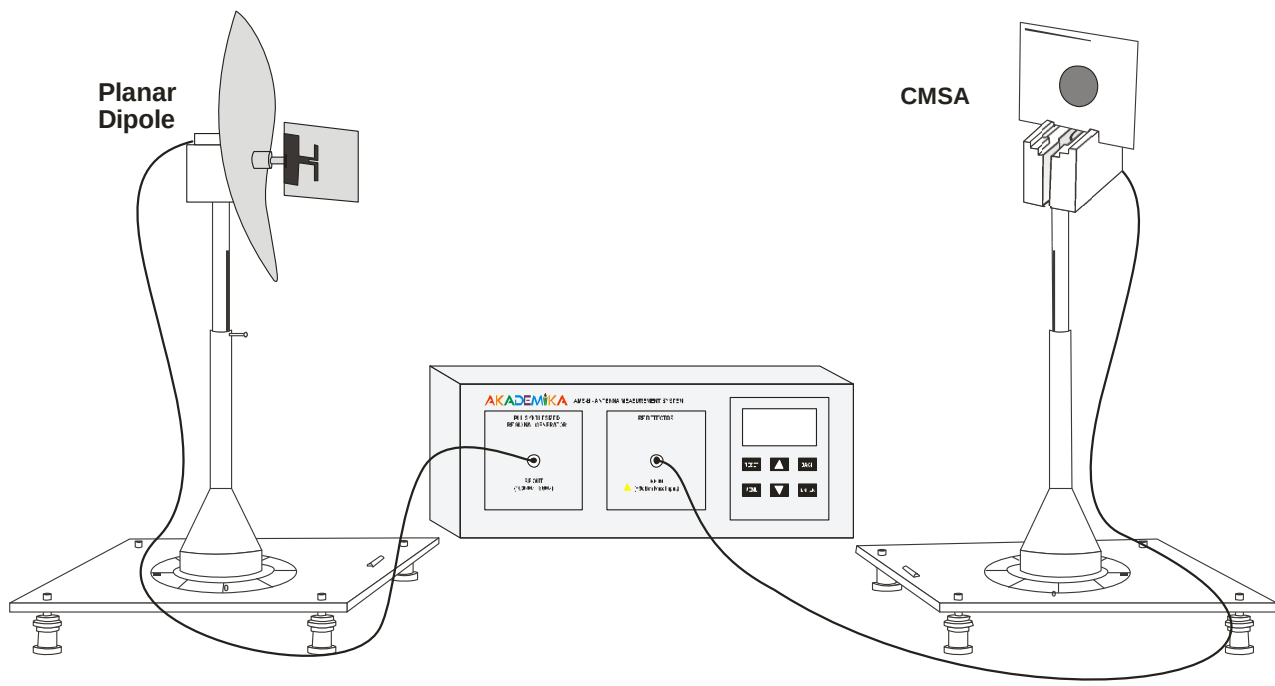
OBJECTIVE:

To plot the radiation pattern of all PLANAR (MICROSTRIP) antenna and observe its parameters.

EQUIPMENT REQUIRED:

TX Antenna	RX antenna	Transmitter input	Receiver output	CABLE
PLANAR DIPOLE	Any PLANAR antenna	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SETUP ARRANGEMENT

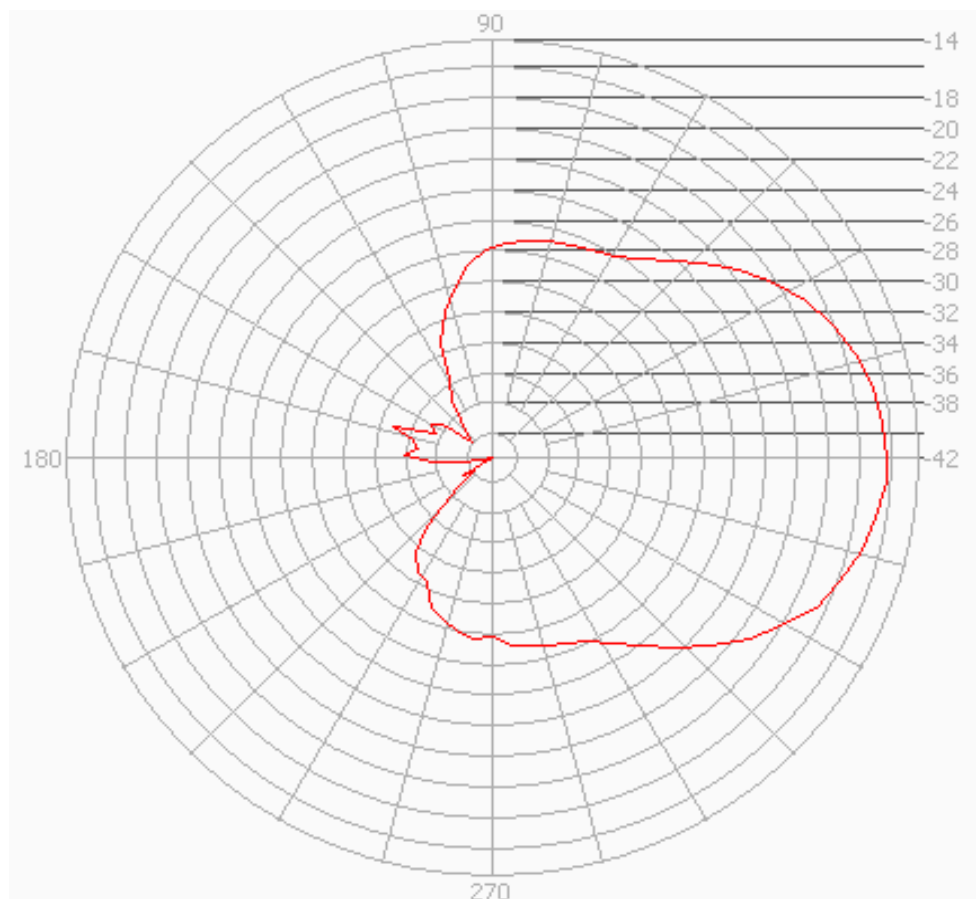


PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 1meter, consult theory for details.
3. Turn ON the module.
4. Open the AMS-B.exe file, select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in FAR FIELD PATTERN, select CO-POLARIZATION, select RX antenna then click on START
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. Now repeat for CROSS-POLARIZATION of antenna and observe the plot.

EXAMPLE 1

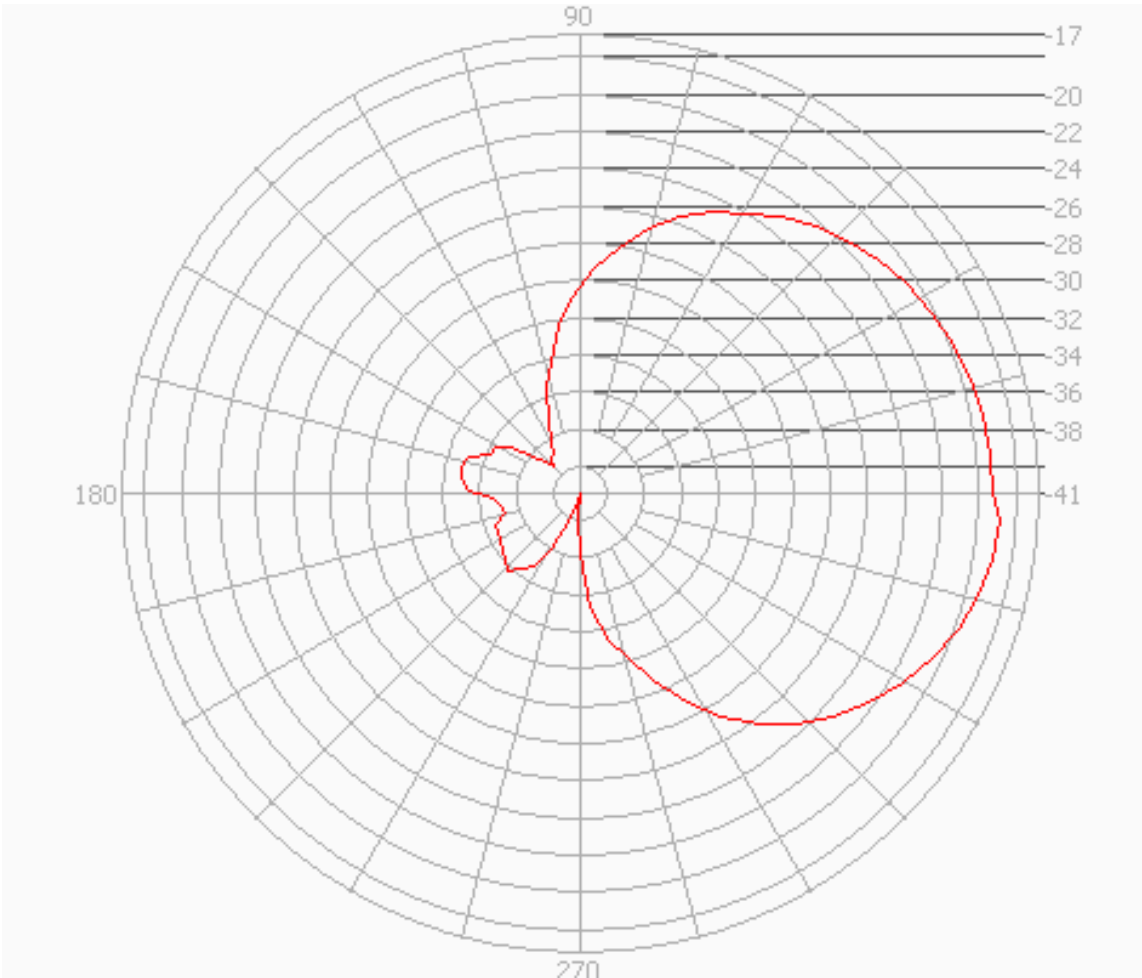
TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Planar Dipole	2 X 1 Array	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN:

EXAMPLE 2

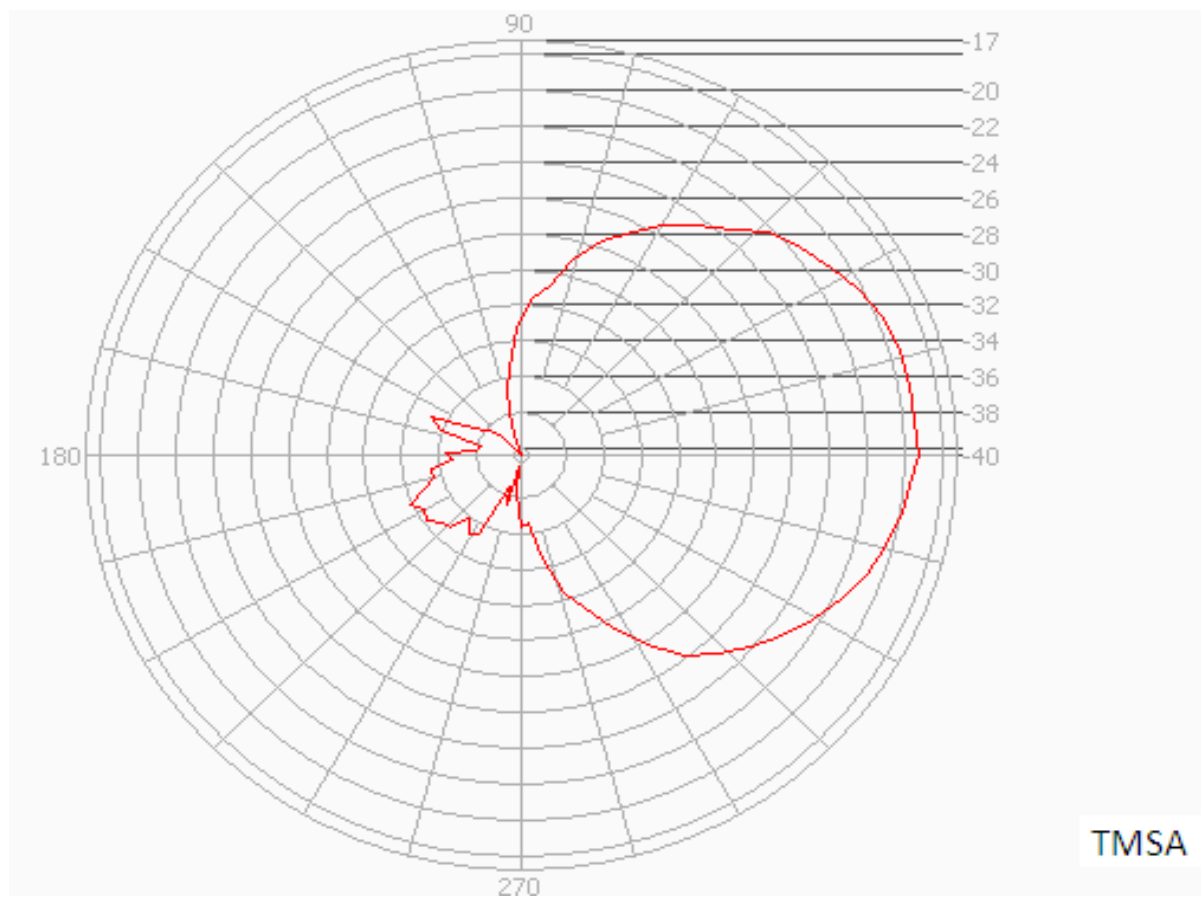
TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Planar Dipole	Insert feed	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN



EXAMPLE 3

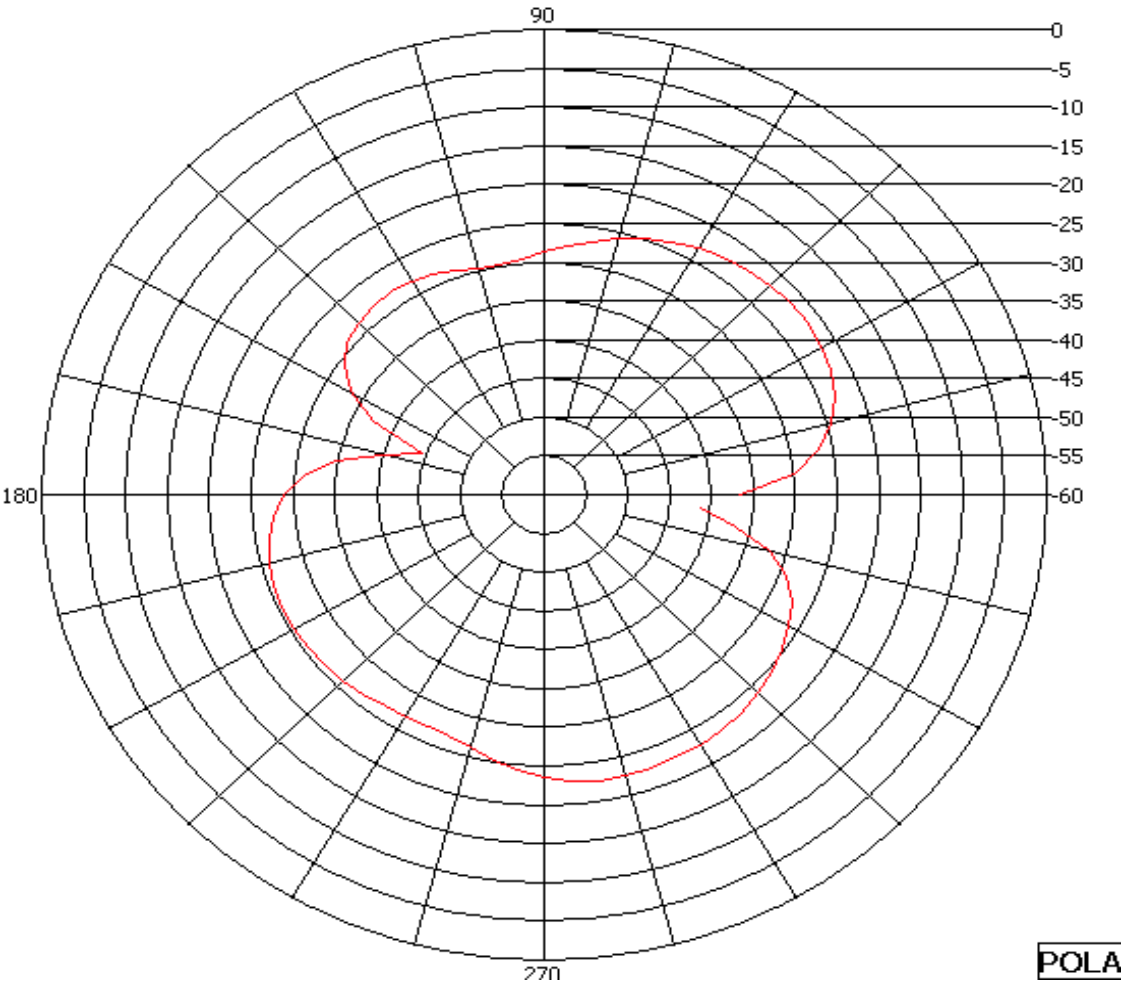
TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Planner Dipole	TMSA	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN

EXAMPLE 4

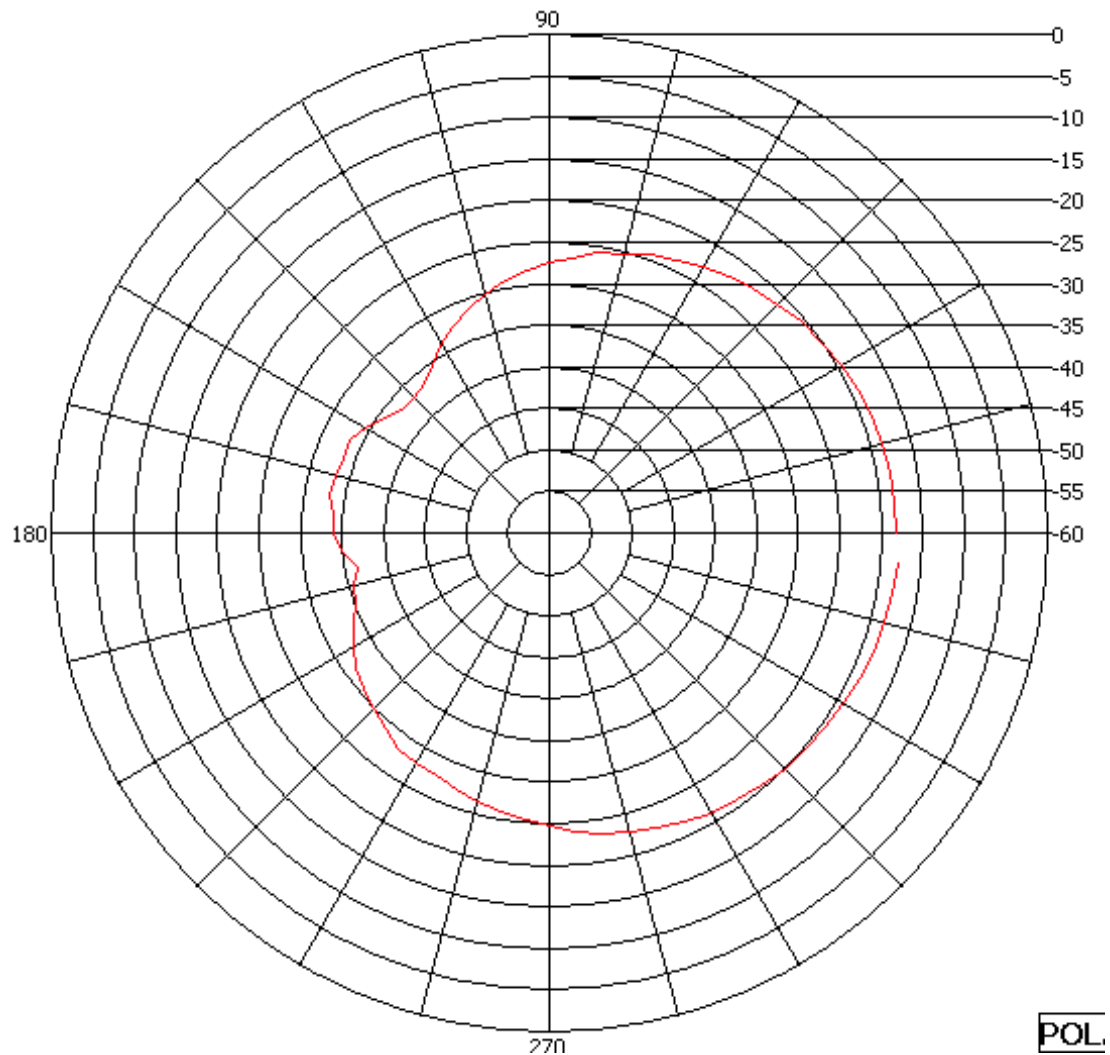
TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Planner Dipole	Angular ring	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN



EXAMPLE 5

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Planner Dipole	CMSA	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2



EXPERIMENT NO. 9.2

DIFFERENT FEEDING TECHNIQUES IN PLANAR (MICROSTRIP) ANTENNA.

OBJECTIVE

Effect of different feeding techniques in PLANAR (Microstrip) antenna.

EQUIPMENT REQUIRED:

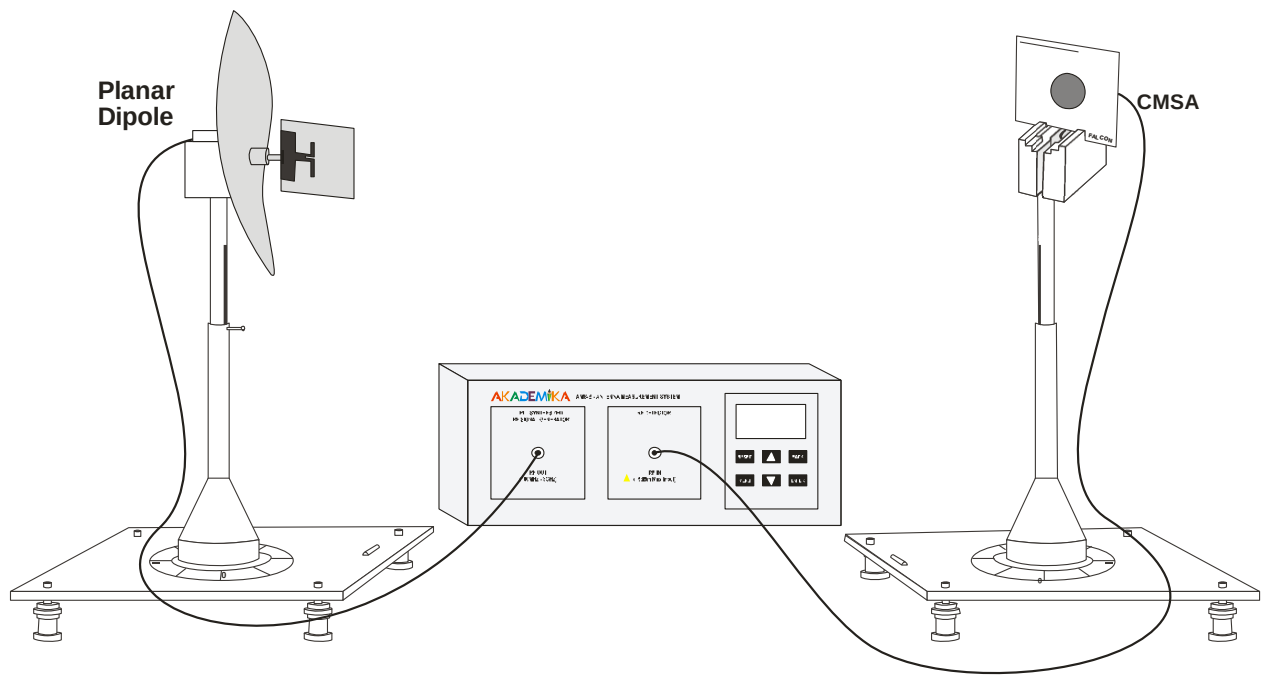
TX Antenna	RX Antenna -Shapes	Transmitter input	Receiver output	CABLE
PLANAR DIPOLE	Any PLANAR (Microstrip) antenna With feedings. 1. Co-axial 2. Insert feed 3. Electromagnetically coupled feed.	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

PROCEDURE:

1. Set up the experiment as per shown in figure above connect using Co-axial feed.
2. Set the distance between the antennas to be around 1meter, consult theory for details.
3. Turn ON the module.
4. Open the AMS-B.exe files; select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in FAR FIELD PATTERN, select CO-POLARIZATION, select RX antenna then click on START.
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. Similarly, repeat the experiment for other feeding techniques i.e. insert feed and electromagnetically coupled feed and observe the effect on received power.

EXAMPLE 1: CO-AXIAL FEED

TX antenna	RX Antenna	Transmitter input	Receiver output	CABLE
Planner Dipole	CMSA –Coaxial feed	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SETUP ARRANGEMENT

RADIATION PATTERN: